HEAL-DP: Homomorphic Encryption Aggregated Learning with Differential Privacy

Unlocking the value of health data without compromising privacy

Akasha Shafiq
University College Cork, Ireland
School of Computer Science
a.shafiq@umail.ucc.ie

Jonathan Pearson
NHS England
Data Science, Transformation Directorate
jonathanpearson@nhs.net

Executive Summary

This project presents HEAL-DP: Homomorphic Encryption Aggregated Learning with Differential Privacy, an implemented and experimentally evaluated framework demonstrating how complementary Privacy-Enhancing Technologies (PETs) can address distinct privacy requirements in healthcare analytics. The framework integrates Differential Privacy (DP), Homomorphic Encryption (HE), Federated Learning (FL), and operational privacy governance to examine privacy protection alongside analytical utility, computational costs, and trust assumptions.

Healthcare organisations increasingly rely on patient data to support clinical research, population health analysis, service planning, and predictive modelling. However, the sensitive nature of healthcare information creates challenges when data must be processed or analysed across organisational boundaries. Conventional data protection measures, including de-identification and access controls, do not independently address all forms of information disclosure that may arise during computation or through analytical outputs. HEAL-DP investigates these challenges across two settings: distributed statistical analysis and collaborative predictive modelling, using MIMIC-IV data partitioned across five simulated hospital nodes.

The framework was implemented and evaluated across three complementary components:

- **Differentially Private Statistical Analytics:** Predefined aggregate queries were implemented with patient-level contribution constraints, sensitivity-calibrated noise, and cumulative privacy-budget enforcement. Evaluation examined statistical utility, population heterogeneity, and disclosure risks through controlled inference and differencing attacks. The results highlighted challenges for small patient populations and the importance of accounting for repeated releases and adversarial background knowledge.
- **Homomorphic Encryption-Enabled Analytics:** HE was integrated to protect intermediate aggregates during distributed computation, with DP noise incorporated before decryption. Experimental evaluation validated the correctness of supported encrypted operations and assessed the additional computational overhead associated with encrypted processing.
- **Differentially Private Federated Learning:** Patient-level DP was investigated for predicting prolonged intensive care unit stays without centrally pooling patient records. Several privacy configurations preserved predictive discrimination relatively close to non-private reference models. However, the results also identified differences in predictive performance across patient subgroups and simulated hospitals highlighted the importance of evaluating performance beyond aggregate predictive accuracy.

Collectively, the findings demonstrate the practical integration of complementary PETs within controlled healthcare analytical workflows. Rather than prescribing a universal privacy configuration, HEAL-DP provides an experimental basis for assessing and implementing privacy guarantees, as well as their analytical usefulness, operational responsibilities, and residual risks. Further, security, institutional, and operational validation remains necessary before deployment in real-world healthcare environments.

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Glossary of Terms and Abbreviations

Term	Definition
Adjacency	A definition of neighbouring datasets, specifying how the data of one individual may differ between them. It determines the scope of a DP guarantee.
Attribute Inference	An attack in which an adversary attempts to infer a sensitive attribute of an individual from available information, including statistical outputs.
BFV (Brakerski-Fan-Vercauteren)	A homomorphic encryption scheme supporting exact arithmetic on encrypted integers. Used in this work for counts and histograms.
CDP (Central Differential Privacy)	A DP model in which a trusted processing component computes or receives the necessary statistics and applies privacy-preserving randomisation before releasing results.
CKKS (Cheon-Kim-Kim-Song)	A homomorphic encryption scheme supporting approximate arithmetic on encrypted numerical values. Used for bounded real-valued sums.
Ciphertext	Data represented in encrypted form, requiring the appropriate decryption key to recover the underlying plaintext.
Contribution Bounding	Restricting the amount of information an individual can contribute to an analysis, allowing query sensitivity or training contributions to be controlled.
DP (Differential Privacy)	A mathematical privacy framework that limits how much the distribution of an observable output can change when an individual's data is added, removed, or modified according to a defined neighbouring relation.
DP-SGD (Differentially Private Stochastic Gradient Descent)	A model optimisation approach that uses bounded gradients and calibrated random noise to support differential privacy during training.
DPIA (Data Protection Impact Assessment)	A structured assessment of privacy and data-protection risks, the measures intended to mitigate them, and the residual risks of a processing activity.
EHR (Electronic Health Record)	A digital record containing information associated with an individual's healthcare, including clinical observations, diagnoses, treatments, and outcomes.
FedAvg (Federated Averaging)	A federated optimisation method in which a coordinator aggregates model parameters or updates obtained from participating clients to construct a shared model.
FL (Federated Learning)	A distributed machine-learning approach in which participating data holders contribute to training a shared model without centrally pooling their underlying datasets.
FPR (False Positive Rate)	The proportion of actual negative cases incorrectly classified as positive by a predictive model.
HE (Homomorphic Encryption)	A cryptographic technique that allows supported mathematical operations to be performed on encrypted data without first decrypting it.
Laplace Mechanism	A DP mechanism that adds noise drawn from a Laplace distribution, calibrated according to the query sensitivity and privacy parameter.
LOS (Length of Stay)	The duration of a patient's hospital admission or ICU stay, depending on the analytical task.
MAE (Mean Absolute Error)	The average absolute difference between estimated or released values and their reference values.
MIMIC-IV (Medical Information Mart for Intensive Care IV)	A de-identified clinical research dataset containing hospital and intensive care information, used as the data source for the experiments.
Patient-Level Privacy	A privacy model in which the protected unit is an individual patient, including the eligible records or observations associated with that patient.
PETs (Privacy-Enhancing Technologies)	Methods and technologies designed to reduce exposure, disclosure, or unauthorised use of sensitive information during data processing and analysis.
PR-AUC (Area Under the Precision-Recall Curve)	A measure of predictive discrimination based on the relationship between precision and recall across classification thresholds.
Privacy Accountant	A component or procedure that calculates cumulative privacy expenditure across multiple DP operations or training steps.
Privacy Budget (ε)	A parameter used to quantify the strength of a DP guarantee. Its interpretation depends on the privacy definition, neighbouring relation, and mechanism being evaluated.
Privacy Governance	Policies and operational controls for managing authorised analytical operations, privacy expenditure, access decisions, and accountability.

Term	Definition
Privacy Parameters (ε, δ)	Parameters of approximate differential privacy, where ε bounds the multiplicative change in output probabilities and δ provides an additive allowance. Pure DP uses $\delta = 0$.
Privacy Unit	The entity protected by a DP guarantee, such as a record, patient, or organisation, as determined by the defined adjacency relation.
Query Differencing	An inference technique that compares related statistical results to isolate information about an individual's contribution.
RDP (Rényi Differential Privacy)	A privacy-accounting formalism based on Rényi divergence, used to analyse cumulative privacy expenditure in iterative mechanisms.
ROC-AUC (Area Under the Receiver Operating Characteristic Curve)	A measure of a classifier's ability to distinguish positive and negative cases across classification thresholds.
Sensitivity	The maximum change in a query result between neighbouring datasets under the specified adjacency relation.
Subgroup Disparity	A difference in a performance measure between population groups, such as differences in recall or false-positive rates.
TPR (True Positive Rate)	The proportion of actual positive cases correctly identified by a predictive model; also referred to as recall or sensitivity.
Trust Boundary	A point in a system where information passes between components or entities with different access privileges, responsibilities, or security assumptions.

1 Introduction

Health data constitutes a critical resource for advancing clinical research, service planning, population-level epidemiological analysis, and the development and evaluation of data-driven healthcare systems. Electronic health records (EHRs) can support analyses ranging from descriptive population statistics to investigations of clinical outcomes and health inequalities. However, the same richness of these data that makes it valuable also makes its use privacy sensitive. Clinical datasets may contain detailed demographic, diagnostic, treatment, and outcome information, and consequently require appropriate controls over how data are accessed, processed and released.

Conventional approaches based on de-identification, such as anonymisation, pseudonymisation [1], removal of Personal Identifiable Information (PII), data access controls, and restricted computation environments, are not sufficient for published analytics [2, 3, 4, 5]. In addition, these approaches do not themselves address every form of information disclosure that may emerge throughout the analytical process. Although the mechanisms may successfully restrict direct access to discrete records, sensitive information can nonetheless be deduced from statistical outputs, intermediate computational values, repeated queries, or combinations of released information. Consequently, protecting the sensitive information requires considering the complete analytical workflow rather than only the storage or direct disclosure of raw records.

Privacy-Enhancing Technologies (PETs) provide mechanisms to reduce various forms of information disclosure during data processing and analysis. Healthcare PETs can be organised into two categories - **Algorithmic PETs and Architectural PETs**. The former consists of cryptographic-assisted technologies such as Multi-party computation (MPC) [6], Homomorphic Encryption (HE) [7, 8], Private Set Intersection (PSI) [9], Differential Privacy (DP) [10], Zero-knowledge proof [11] and synthetic data. The second category includes federated learning (FL) [12] and trusted execution environments (TEEs) [13], which primarily limit the information disclosed. These techniques offer different privacy properties and rely on different assumptions. Differential privacy (DP), for instance, provides a formal mechanism to limit the influence of the individual's data on the released statistical results. In contrast, cryptographic techniques such as HE allow computation while the values remain encrypted. Neither property subsumes the other: protecting data during computation does not necessarily protect information revealed by the final result, while protecting a statistical result does not in itself conceal all the values processed during the preceding computation.

This distinction motivates a shift from considering PETs solely as independent privacy mechanisms towards considering complementary PETs that can be incorporated within an end-to-end analytical architecture. Recent work on PET deployment similarly emphasises a problem-oriented view in which techniques are selected according to the exposure surface they are intended to address [14, 15], and wherein the integration of multiple PETs may prove necessary when a given application encompasses diverse privacy considerations. The incorporation of multiple PETs, requires meticulous architectural considerations.

1.1 Form Individual PETs to Complementary Privacy Protection

The deployment of multiple PETs, however, does not inherently yield enhanced privacy assurance: each technique operates under particular conditions, technical specifications, operational expenses, and defined protective parameters. In addition, health analytical systems often require compiling several requirements simultaneously: protecting sensitive patient-level information, enabling useful statistical analysis, controlling information across repeated analysis, maintaining an auditable record of data use, and supporting practical computation performance. These requirements extend beyond the guarantees of any individual privacy mechanism and motivate architectures wherein privacy technologies and administrative governance controls perform complementary roles.

Privacy-enhancing technologies such as homomorphic encryption and differential privacy are often presented as solving the privacy problem outright. In practice they do something narrower and more useful: they change which parties can see what, and they bound what can be inferred from released outputs. Those changes only matter to a data controller once they are expressed in the Data Protection Impact Assessment's (DPIA) [16] own terms: the risks to individuals, their likelihood and severity, the measures that mitigate them, and the risk that remains.

Assessment (DPIA) requires consideration of both risks mitigated by the selected mechanisms and those introduced through their deployment. A comparison of the conventional processing architecture and its PET-enhanced counterpart should therefore identify the original disclosure risks, the extent to which the implemented safeguards address them, any additional vulnerabilities or trust dependencies arising from their integration, and the residual risks that remain under the stated operational assumptions.

For example, incorporating Homomorphic encryption into distributed statistical analytics reduces the exposure of plaintext intermediate values during aggregation but introduces security-critical dependencies on cryptographic key generation, storage, access control, and decryption. Consequently, key compromise or unauthorised use of the decryption

service represents a potential risk introduced by the protection mechanisms themselves. Similarly, differential privacy limits individual-level disclosure through statistical releases, but its effectiveness depends on appropriate mechanism configuration and enforcement of cumulative privacy expenditure. These considerations demonstrate that the adoption of complementary PETs must be assessed not solely by the additional protections they provide, but by their overall effect on privacy risk, introducing newly introduced vulnerabilities and risks that remain insufficiently mitigated.

Without that link, a PETs framework risks being treated as a technical assertion rather than a demonstrable control. Over-reliance on it can also create false assurance.

The clearest way to characterise the framework's value is a comparative DPIA, assessing the same analysis with and without PETs. The baseline is the conventional alternative: pooling data centrally, or a federated aggregate computed in plaintext. Against this, the PETs case sets out for each identified risk which mechanism addresses it, what the privacy claim is, and what residual risk remains. For example, exposure of trust-level counts to an aggregator is mitigated by encryption. Re-identification of individuals through differencing or small cells in the released results is bounded by differential privacy at a stated ε. The comparison should be equally explicit about risks the PETs introduce or leave untouched: key management and collusion between trusts, misconfiguration, the cumulative privacy budget across releases, and the reduced accuracy for small subgroups that noise causes.

We structure the risks to be discussed in the DPIA around:

- Input risk concerns: exposure of identifiable information before or during computation [17, 18]
- Transfer and sharing risks: these risks arise when the information moves between hospitals, devices, research institutes, and cloud servers [19, 20]
- Computation risks: these risks concern access to sensitive values while analysis is being performed [21]
- Model and intermediate output: this concerns information that can be inferred from gradients, model parameters, embeddings, or other intermediate [22, 23, 24] representations when using machine learning
- Output risks: it concerns information that can be inferred from the published statistics, predictions, synthetic data, datasets or trained models [23, 25]

Considering the DPIA lens should shape the PETs architecture and feed into any technical design. This is the surest way of creating a solution which is clear and effective on how it handles the privacy issue.

1.2 Project Aim

The project aims to investigate and develop a framework that enables privacy-preserving healthcare analytics across trusts through two related yet operationally distinct forms of computation: centrally executed statistical analysis and federated model training. The first approach concerns central aggregate analysis. In this setting, authorised statistical queries are evaluated on locally available healthcare data from a single trust. Central differential privacy (CDP) is applied to calibrate the noise prior to releasing the corresponding results. This aims to enable useful aggregated analysis while formally limiting the influence that an individual's record can have on an observable output. While the implementation of the DP mechanism is a necessary component, it does not, on its own, establish a comprehensive operational privacy infrastructure. A deployable analytic server must specify permissible query categories, compute the associated privacy expenditure, monitor aggregate privacy depletion, deny requests that exceed available budget allocations and perform privacy-budget management over the system lifecycle.

The implementation of DP thus constrains the information disclosed through the observable output; it does not provide confidentiality for the values processed prior to release. The framework, therefore, investigates Homomorphic encryption as a complementary protection mechanism within the centralised workflow.

The second analytic setting considers an environment in which healthcare data is distributed across different organisations, and central pooling of patient-level records is undesirable. Federated learning provides an alternative computational approach in which participating hospitals perform local model training and contribute information required to construct a shared model while retaining their original records. Consequently, the framework adopts a complementary rather than a substitutive view of PETs. Different privacy methodologies are assigned to specific exposure surfaces depending on the underlying processing scenario.

Therefore, for the centralised analytics pathway, the framework combines

- central differential privacy to protect released statistical results;
- privacy budget accounting to control cumulative privacy expenditure;

- query governance mechanism to enforce privacy policies;
- homomorphic encryption to protect the results within the computation and release pathway;
- auditing to provide traceability if privacy-relevant operations.

For the distributed learning pathway, the framework combines:

- federated learning to retain patient-level dataset at participating hospitals;
- DP to limit individual influence on the information revealed through iterative model training; and
- privacy accounting to quantify and manage the cumulative privacy expenditure across the learning process.

Furthermore, both analytical settings were evaluated to determine how privacy configurations affect the utility of central statistical results, the model's predicted performance, and the computational costs introduced by privacy-preserving mechanisms.

1.3 Project Scope and Outputs

The project provides an implemented, end-to-end experimental framework for investigating interactions among statistical privacy, distributed learning, cryptographic protection, and operational privacy governance in a healthcare setting. The two distinct analytical settings enable the study of different combinations based on the processing architecture and corresponding privacy risks. The framework is positioned with a risk-led perspective established by a DPIA [16]. Accordingly, individual mechanisms and PET combinations are considered in relation to the exposures they are intended to mitigate, their associated trust assumptions, and the residual risks that remain after safeguards are applied. The framework should therefore be understood as an investigation into how technical safeguards can support privacy-by-design within defined analytical workflows, rather than as a claim that the deployment of a particular PET or combination is sufficient by itself to satisfy all data-protection requirements.

Beyond evaluating the technical properties of individual PETs, the project aims to demonstrate their practical application within healthcare analytical workflows. In particular, it seeks to bridge the gap between the theoretical foundations of Differential Privacy and its operational deployment by illustrating how privacy mechanisms can be configured, integrated, and governed alongside analytical tasks. Using an experimental framework, the work provides concrete examples of how to implement privacy-preserving statistical analysis and distributed model training, while examining the implications for analytical utility, computational performance, and privacy protection.

2 Background and Related Work

Privacy-Enhancing Technologies (PETs) can act on different elements of a processing workflow. Some mechanisms constrain the information that can be inferred from an output, while others reduce exposure during computation or change where data are processed. This chapter focuses on the technologies relevant to the framework developed in this project. Differential privacy is considered a formal disclosure-control mechanism for both statistical releases and model training; Homomorphic Encryption provides protection during computation, and Federated Learning enables collaborative model training without requiring central pooling of raw records.

Adjacency (Privacy Unit). Two datasets are said to be adjacent if they differ in the data of exactly one individual. This could be achieved by adding or removing an individual, or by modifying an individual's data. It defines the smallest possible difference between datasets.

2.1 Differential Privacy (DP)

DP provides a mathematical guarantee that limits the influence that the inclusion or exclusion of an individual can have on an observable output, achieved by adding carefully calibrated noise [10, 26]. Let D and D' denote two neighbouring datasets differing in one record, and let M denote a randomised mechanism. A mechanism satisfies (ε, δ)-DP if for every measurable set of outputs (S)

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\epsilon} \Pr[\mathcal{M}(D') \in S] + \delta.$$

The parameter ε, also referred to as privacy budget, is responsible for the addition of noise; it controls the maximum multiplicative change in the probability of an observable output between neighbouring datasets, while δ permits a bounded probability of deviation from the pure ε-DP guarantee. Smaller values of ε generally correspond to a stronger privacy (more noise), and larger values of ε correspond to more utility (less noise).

Post-Processing. It provides a guarantee that arbitrary computational operations performed on the output of a differentially private mechanism maintain the established privacy guarantees, thereby eliminating any possibility of compromising the privacy protections that the mechanism has provided

Consequently, privacy accounting plays an important role in connecting the mathematical DP mechanism to the systems in which it is deployed.

DP is particularly relevant to healthcare because it protects against disclosure through an observable output rather than solely by removing identifiers. However, DP is not a single deployment architecture; the point at which randomisation occurs determines the associated trust assumptions and the type of processing it supports. There are two modes of differential privacy **Central Differential Privacy (CDP) and Local Differential Privacy (LDP)**.

2.1.1 Central Differential Privacy for Statistical Queries

Under CDP, a trusted curator has access to the original data and is responsible for applying randomised noise before publishing the results. This model is appropriate when the curator lies within the trusted processing boundary, and the principal privacy concern is disclosure through the analytical outputs. For a query f, the magnitude of perturbation required depends on its sensitivity, describing the maximum change in the output based on the adjacency unit. The global sensitivity can be expressed as

$$\Delta f = \max_{D \sim D'} |f(D) - f(D')|_1 ,$$

where $(D \sim D')$ denotes neighbouring datasets. for pure ε-DP, the Laplace mechanism releases

$$f(D) + \mathrm{Lap}\left(\frac{\Delta f}{\epsilon}\right).$$

The **magnitude of the added noise is therefore determined jointly by the query's sensitivity and the allocated privacy budget**. For a fixed sensitivity, smaller values of ε require greater perturbation, while larger ε permits greater statistical accuracy. For a mechanism satisfying approximate differential privacy, Gaussian noise is calibrated based on the ε, δ parameters, the sensitivity, and the accounting framework used.

2.2 Homomorphic Encryption (HE)

Homomorphic encryption addresses a different privacy objective by allowing supported operations to be performed directly over ciphertexts. HE addresses confidentiality during data processing by allowing supported operations to be evaluated directly over ciphertexts. This property is particularly relevant to collaborative healthcare analytics, where multiple institutions may contribute statistical information without disclosing their individual plaintext contribution to a central coordinator.

An HE scheme generally comprises four operations: key generation, encryption, homomorphic evaluation and decryption. Key generation produces a public key pk and a secret key sk. The public key is used to encrypt plaintext, while the secret key is used to decrypt the resulting ciphertexts. The evaluation operations apply a supported function f to encrypted inputs, producing a ciphertext that represents the corresponding computation over the underlying plaintexts. This property can be expressed as

$$\mathrm{Dec}_{sk}\left(\mathrm{Eval}\left(f, \mathrm{Enc}_{pk}(m_1), \ldots, \mathrm{Enc}_{pk}(m_n)\right)\right) \approx f(m_1, \ldots, m_n), \tag{1}$$

where $m_1, \ldots, m_n$ denote plaintext inputs. The relationship is exact for schemes supporting exact arithmetic, subject to their encoding and parameter constraints, whereas approximate schemes introduce numerical approximation during computation.

HE schemes differ in the types and complexity of operations they support. Partially Homomorphic Encryption (PHE) permits a restricted set of operations, such as repeated encrypted addition or multiplication. Somewhat Homomorphic Encryption (SWHE) supports a limited sequence of arithmetic operations, while Fully Homomorphic Encryption (FHE) permits the evaluation of arbitrary computable functions through suitable cryptographic constructions. The choice of scheme, therefore, depends on the required computation, supported numerical representation, and associated computation overhead.

In statistical analytics, additive homomorphic operations are particularly relevant because many aggregate queries can be expressed in terms of counts, sums, and related sufficient statistics. The Brakerski-Fan-Vercauteren (BFV)

scheme supports exact arithmetic over encoded integers and is consequently suitable for operations involving counts and categorical distributions. In contrast, the Cheon-Kim-Kim-Song (CKKS) scheme supports approximate arithmetic over encoded real-valued data, making it suitable for bounded numerical sums. Statistical measures such as means can be constructed from encrypted sums and counts, with the final calculation performed after the required protected aggregate has been obtained. These characteristics motivate the use of BFV and CKKS in the statistical analytics pathway considered in this work.

Although HE reduces plaintext exposure during computation, it does not inherently prevent information disclosure from decrypted analytical results. Its confidentiality guarantees also depend on the security of the underlying cryptographic scheme, appropriate parameter selection, and protection of the secret decryption key. Furthermore, encrypted computation introduces computational and communication overhead, while approximate arithmetic may affect numerical fidelity. Consequently, HE and DP address distinct yet complementary aspects of privacy protection: HE restricts access to sensitive intermediate values, whereas DP limits disclosure of individual-level information through the released analytical results.

2.3 Federated Learning

Federated Learning (FL) [12] changes the architecture of machine-learning training by allowing multiple data holders to contribute to building a shared model while retaining their underlying datasets locally. Consider k participating nodes and optimise a local objective $(F_k(w))$. The corresponding global optimisation objective may be represented as

$$\sum_{k=1}^{K} \frac{n_k}{n} F_k(w), \qquad n = \sum_{k=1}^{K} n_k.$$

In a conventional Federated Averaging configuration, the server distributes the current global model to participating nodes. Each node performs a local optimisation using its own data and subsequently returns an updated representation of the model. The server aggregates these contributions to construct the next global model.

2.3.1 Differential Privacy in Federated Learning

Combining DP with federated learning [27, 28] addresses two distinct dimensions of privacy protection. Federated Learning reduces the need to centrally collect raw records, whereas DP provides a formal mechanism to limit the influence of protected data on information exposed during the learning procedure. The resulting privacy guarantees depend on the neighbouring relation and the corresponding unit of privacy. The combination of FL and DP exemplifies layered privacy protection. FL reduces the risk of data transfer and the centralisation of patients' records, but inference risks may remain through information exchanges during learning. DP can therefore be introduced at the training level to reduce the influence of individual data on those observable quantities.

2.4 Privacy Accounting, Governance and Operation Enforcement

Formal privacy guarantees exist within larger operational systems. DP guarantees depend not only on correct noise calibration but also on how privacy-consuming operations are managed over time. A privacy accountant maintains information on cumulative privacy expenditure. In an interactive statistical service, this enables the system to determine whether executing a new query is compatible with the remaining privacy budget. In iterative model training, accounting instead combines the privacy effects of repeated optimisation steps and sampling operations.

Privacy accounting should be distinguished from privacy governance. Accounting quantifies privacy expenditure, whereas governance determines whether execution of a new query is compatible with the remaining how this information affects system behaviour. A system may, for example, impose an overall privacy limit, restrict operations that consume a substantial fraction of the available budget, require additional confirmation for particular requests or associate expenditure. The distinction between calculation and enforcement is significant.

Consequently, privacy accounting forms part of the operation architecture surrounding DP rather than merely an auxiliary calculation. The system must maintain consistency between the requested privacy parameters, authorised expenditure, mechanism execution and recorded privacy state. The framework developed later in this report operationalises this relationship in the central analytical pathway. Privacy expenditure is evaluated before execution, while policy controls determine whether an operation may proceed.

2.5 Combining PETs for Layered Privacy Protection

The preceding technologies address different privacy properties and different stages of the analytical lifecycle. Their combination is therefore best considered an architectural design problem. The healthcare literature provides several examples of such layered protection [15, 14]. Sheller et al. [20] demonstrated the feasibility of multi-institutional model training without centrally sharing patient data. Extending this approach, Sadilek et al. [29] integrated DP with FL across clinical and epidemiological studies, examining privacy protection alongside model utility. Similarly, Shiri et al. [30] investigated differentially private federated learning for COVID-19 prognosis using multi-institutional chest CT data. More recently, Shahid et al. [31] combined federated learning, differential privacy, homomorphic encryption and secure aggregation for chronic disease prediction using the MIMIC-III dataset. Collectively, these studies illustrate how data localisation, formal disclosure protection, and cryptographic confidentiality can address distinct privacy requirements within healthcare analytical workflows.

The use of multiple PETs should not, however, be interpreted as automatically producing a stronger overall privacy guarantee. Individual PETs have different assumptions and protection boundaries, and their combination can introduce additional computational, communication, configuration and governance requirements. Algorithmic PETs may affect analytical utility, while cryptographic mechanisms can increase computation cost. A complex combination can also introduce additional trust relations.

Therefore, a combined architecture should be assessed as actually deployed rather than assuming that the theoretical properties of individual components automatically compose into an adequate system-level guarantee. This leads to the risk-oriented interpretation of PET composition. The question is not how many privacy technologies are present within a system but whether each mechanism corresponds to an identified exposure surface and whether residual risks remain after that protection has been introduced. This layered interpretation of privacy provides the conceptual basis for the framework developed in this project.

2.6 HEAL-DP Framework

The framework developed in this project builds upon established Differential Privacy (DP), Homomorphic Encryption (HE) and Federated Learning (FL) techniques to investigate their complementary application in privacy-preserving healthcare analytics. It considered two analytical settings: centralised statistical analysis integrating DP, privacy-budget governance, and HE-enabled computation and distributed model training combining FL with DP. Guided by a risk-informed approach, the framework focuses on integrating and experimentally evaluating these mechanisms, examining their respective privacy protections, analytical utility, and computational trade-offs.

HEAL-DP is developed for NHS England, intended to support healthcare organisations in deriving and sharing analytical insights from sensitive health data while limiting unnecessary exposure of patient-level information. Through its implementation and experimental evaluation, this work demonstrates the practical feasibility of integrating complementary PETs within controlled statistical analyses and collaborative learning Workflows, while identifying the privacy, utility and operational considerations relevant to their adoption in healthcare environments.

3 Privacy Requirements and Threat Model

3.1 Healthcare Processing Scenarios and Data Flows

HEAL-DP considers two healthcare analytical scenarios: distributed statistical analysis and collaborative machine learning. Both cases are evaluated using healthcare data derived from the MIMIC-IV [32] dataset and organised across five simulated hospital nodes. Although the underlying data originates from a common research dataset, the partitioned environment represents a multi-institutional setting in which participating healthcare organisations retain control over their respective patient records.

The two scenarios differ in their computational objectives and the information exchanged between participating entities. The statistical analysis pathway supports the release of aggregate information through a controlled query interface, whereas the federated learning pathway supports collaborative predictive modelling without centralising the underlying training data. These distinctions determine where sensitive information may become observable and which privacy safeguards are required.

3.1.1 Distributed Statistical Analytics

The first scenario considers a healthcare analytics environment in which multiple hospitals contribute to statistical analysis over their respective patient populations. Such an arrangement is relevant to healthcare organisations seeking

to obtain population-level insights, including patient demographics, admission patterns, and clinical outcomes, without consolidating individual-level records into a shared analytical repository.

The experimental environment comprised five simulated hospital nodes, each maintaining a separate partition of the healthcare dataset. An analyst interacts with the system through a central query interface and submits a request corresponding to a predefined statistical operation. The coordinating component distributes the relevant computation to participating nodes, where local aggregate statistics are computed. These contributions are subsequently combined to obtain the requested population-level statistics.

Under Central Differential Privacy (CDP) configuration, the aggregated statistics are processed by a DP mechanism before being released to the analysts. Although the underlying patient remains distributed, the privacy mechanism operates at the aggregate release stage. This distinction is significant because the use of CDP does not require raw records to be stored centrally; rather, it assumes a trusted processing boundary within which the required computation and randomisation are performed.

The framework additionally considers Homomorphic Encryption (HE)-enabled pathway to reduce the exposure of intermediate aggregate values. In this configuration, participating nodes encrypt their local statistical contribution before transmission. The coordinator combines the encrypted contributions without possessing the corresponding decryption key. DP remains responsible for limiting information disclosed through the final analytical output. Both configurations operate within a controlled query environment in which privacy expenditure is assessed before execution. HEAL-DP associates requests with the defined privacy costs and maintains cumulative accounting and audit information, allowing repeated statistical interactions to be governed under an established privacy policy.

The resulting processing workflow therefore separates three distinct concerns: **local access to patient-level information, exposure of intermediate aggregate values, and disclosures through released statistics.** These concerns provides the basis for identifying the relevant trust boundaries and privacy requirements in subsequent chapter.

3.2 Federated Healthcare Model Training

The second scenario addresses collaborative predictive modelling across distributed healthcare data holders. In contrast to statistical query processing, the objective is to train a shared predictive model using information available at participating hospital nodes while avoiding the transfer of complete patient-level training datasets to a central repository.

HEAL-DP evaluates this setting in a Federated Learning (FL) environment using patient-level data partitions. The analytical task concerns predicting prolonged intensive care unit length of stay using clinical and demographic predictors.

During training, the coordinator distributes model parameters to the selected participants. Each participant updates the model using its local dataset and returns the result training information for aggregation. This process is repeated across multiple communication rounds, allowing the global model to incorporate information from the distributed populations without requiring central access to the underlying patient records.

Although this arrangement reduces the movement of raw clinical information, the exchanged model updates remain a potential source of privacy exposure. As a result, HEAL-DP incorporates differential privacy into the federated training procedure to bound the influence of individual patients on the learning process. Privacy accounting considers the cumulative effect of repeated optimisation steps and training rounds under the specified privacy configuration. The federated scenario, therefore, introduces privacy requirements that differ from those of aggregate statistical release. In particular, the observable information includes model updates, aggregated parameters, and the resulting predictive model, while the privacy cost is associated with an iterative learning procedure rather than a sequence of independent statistical queries.

The distinction between these processing scenarios informs the subsequent threat model: the statistical pathway requires protection of intermediate computations and released aggregates, whereas the federated pathway additionally requires consideration of information leakage through collaborative model training. The relevant actors' protected assets and trust boundaries are identified in the following section.

3.3 Privacy Risk Mitigation and PET Selection

Each PET addresses a particular privacy concern as highlighted in Table 2. The selection of the privacy-enhancing mechanism with the proposed framework is guided by the processing requirements and exposure points identified in the preceding assessment. The used privacy mechanisms were selected because their respective protection properties correspond to distinct requirements of the statistical and distributed learning workflows. Other PETs offer alternative protection models but introduce different trust assumptions, operational requirements, or computational trade-offs.

The selection therefore reflects the suitability of each mechanism for the defined experimental scenarios rather than a general preference for one privacy technology over another.

Table 2: Privacy Guarantees of Algorithmic and Architectural PETs

PET	Primary Protection	Healthcare Use-case	Limitation
Homomorphic Encryption	Data during computation	Cloud computation on encrypted clinical data	Does not provide a guarantee of protection of released data
Private Set Interaction	Private record matching	Cohort matching hospitals	Reveals the intended intersection
Secure Multi-party Computation	Distributed Private Computation	Joint statistics across hospitals	Computational and communicational cost
Zero-knowledge Proofs	Verification without disclosure	Verifying a truth without revealing the underlying medical information	Application-specific and computational complexity
Trusted Execution Environment	Isolated secure computation	Protected health data computation in cloud infrastructure	Hardware and trust vulnerabilities
Federated Learning	Distributed learning with data localisation	Multi hospital models training	Models may learn information
Synthetic Data	Reduced direct use of original data	Research data sharing	Not inherently privacy preserving
Differential Privacy	Inference or output privacy	Statistics, synthetic data generation, ML models	Privacy vs utility trade-off and parameter selection

3.4 System Actors, Protected Assets and Trust Boundaries

The privacy properties of a healthcare analytical system depend not only on the mechanism employed but also on the entities responsible for processing information and the extent of their access to sensitive data. The proposed framework separates data ownership, computational coordination, privacy enforcement and result disclosure into distinct functional roles.

3.4.1 System Actors and Responsibilities

Across statistical analytics and federated learning workflows, participating entities have specific responsibilities and trust assumptions determined by their analytical functions, the information exchanged, and the privacy mechanisms in place. Although evaluated within a simulated multi-hospital environment, these entities represent functions relevant to collaborative healthcare analytics.

Hospital Nodes (Data holders). Five simulated hospital nodes maintain separate partitions of the MIMIC-IV derived dataset. Each node is responsible for processing its local patient records and producing the information required by the requested analytical operation. In the statistical pathway, nodes compute local aggregate contributions, whereas in the federated pathway, they perform local model training. Patient-level records remain within their respective data-holder environments during these operations.

Analytical Coordinator. The coordinator manages incoming statistical requests and orchestrates their execution across participating hospitals. Its responsibilities include query validation, distribution of approved operations, aggregation of local contributions, and coordination with privacy-governance and release outputs. The information accessible to the coordinator depends on the processing configuration. In the plaintext CDP scenario, it can observe aggregate values required for query processing, whereas the HE-enabled pathway restricts its access to encrypted contributions and intermediate ciphertexts.

Privacy-governance and Accounting. These components evaluate the privacy expenditure associated with analytical operations and determine whether execution is permitted under the configured privacy policy. Their responsibilities include maintaining the cumulative budget state, enforcing expenditure limits, managing high-cost requests, and recording privacy-relevant execution decisions. Their principal function is to preserve the integrity and consistency of privacy enforcement across repeated analytical interactions.

Cryptographic and Release Services. The HE-enabled configuration separates encrypted aggregate processing from the operation required to produce a releasable result. Participating nodes encrypt their statistical contributions using the public-key context, while the coordinator performs a supported aggregate operation on the ciphertext. A separate release service maintains the corresponding decryption capability. In the encrypted-noise configuration, noise service generates and encrypts the required DP noise before the resulting ciphertext is delivered for decryption and release. This separation reduces the information accessible to the coordinating component.

Federated Learning Coordinator. Within the distributed learning pathway, the federated coordinator manages model initialisation, participant coordination, training rounds, and aggregation of model updates. Participating nodes retain their training data locally and communicate the information required to update the global model. Unlike the statistical query coordinator, the federated coordinator operates on model-related information rather than patient-level aggregate query results.

Analysts and Authorised Users. Analysts represent the recipient of statistical outputs or trained models. Their access is restricted to information made available through the corresponding analytical interface. They do not require direct access to the underlying patient-level datasets to obtain aggregate statistical results or use the resulting predictive models.

3.4.2 Protected Information and Assets

HEAL-DP processes several categories of information that differ in sensitivity, accessibility, and their potential to disclose patient information. These include primary healthcare records, derived statistical values, model training information, cryptographic material, and privacy governance records.

Patient-level healthcare records constitutes the primary protected asset. The data include demographics and clinical attributes associated with individual patients and their hospital encounters. Such information may support statistical analysis or predictive modelling, but can also expose sensitive patient characteristics if accessed or disclosed outside the processing environment. The experimental data are therefore partitioned across hospital nodes, with patient identity used to define contribution boundaries for privacy-preserving operation.

Local and global statistical aggregates represent a second category of protected information. Although these values are derived from multiple records, exact counts, sums, or subgroup statistics may reveal sensitive characteristics, particularly when contributions are small or when multiple related queries are combined. Their confidentiality and disclosure requirements differ depending on whether they remain within a trusted computation boundary, are processed as ciphertexts, or are released to an analyst.

Model Parameters and training updates constitute an additional category of information within the federated pathway. These values are produced through local optimisation and may retain information about the underlying training population. Consequently, the absence of direct patient record transmission does not imply that the communicated model is free from privacy risk. The relevant protection requirement extends to observable training contributions and the resulting global model.

Cryptographic keys and encryption contexts are security assets in the HE-enabled pathway. Public encryption contexts enable participating components to produce and manipulate ciphertexts, whereas access to the corresponding secret key permits decryption. The confidentiality of the secret key and its separation from components processing encrypted aggregates are therefore central to the intended cryptographic trust boundary.

Lastly, **privacy budget records, query metadata and audit information** are essential to the integrity and accountability of the analytical process. These records represent privacy allocation, execution decisions, cumulative expenditure and relevant processing outcomes. Their integrity is particularly important because inaccurate accounting or inconsistent state changes may undermine the enforcement of the configured privacy policy.

3.4.3 Trust Boundaries and Information Exposure

A trust boundary identifies the point at which information passes between entities with different access privileges, responsibilities, or security assumptions. Within the proposed framework, these boundaries vary depending on whether statistical computation is performed in plaintext or under HE, and whether the analytical operation involves aggregate query processing or federated model training.

Statistical Computation and Release Boundary. In the CDP-only scenario, local statistics are aggregated using the trusted analytical setting. The DP mechanism subsequently perturbs the computed statistics before they are disclosed to the analysts. The corresponding assumption is that components authorised to observe the exact aggregate operate within

the trusted processing boundary and do not disclose unprotected results. DP limits information disclosure through released output, but does not remove this trust requirement. The HE-enabled pathway modifies this boundary by encrypting local statistical contributions before aggregation. Therefore, the confidentiality of intermediate aggregates no longer depends on granting the coordinator direct access to their exact values.

Federated training Boundary. In this setting, the principal boundary separates patient-level datasets retained at participating hospitals from model information communicated to the federated coordinator. The coordinator receives the training contribution required for global optimisation. However, the model updates and parameters remain observable information and may contain data-dependent characteristics. Differential privacy is consequently incorporated into the learning process to constrain the influence of protected patient information under the specified privacy model. The resulting guarantee depends on the unit of privacy, the placement of randomisation and the cumulative accounting procedure.

Governance and Disclosure Boundary. Both the setting require distinction between information processed internally and information permitted to leave the system. In a statistical setting, privacy budget checks and execution controls govern whether a DP result may be released. Audit records provide traceability of the corresponding decision and expenditure. In a federated environment, the observable information includes communicated training contributions and the final model, requiring privacy protection to be evaluated across the full learning process.

These trust boundaries establish the conditions under which the proposed mechanisms are intended to operate. In particular, encryption does not eliminate the need for trusted key management, federated learning does not conceal all information contained in model updates, and DP does not independently guarantee the confidentiality of the values processed within a trusted environment. The adversarial capabilities and privacy risks associated with these boundaries are examined in the following chapters.

3.5 Threat Model and Assessment Scope

The threat assessment primarily considers an honest-but-curious adversarial model that follows the prescribed protocols but may attempt to infer sensitive information from observable outputs or intermediate computations. Potential adversaries include authorised analysts examining statistics, coordinating components accessing computational outputs, or participants observing federated model updates. Adversaries may also possess auxiliary information and correlate observations across multiple authorised interactions.

The assumed adversarial capabilities are constrained by the trust boundaries defined in Section 3.4. The analytical coordinator is assumed not to possess any private context, while participating hospital nodes are assumed to maintain appropriate controls over patient-level records. The visibility of model updates and intermediate statistical values depends on the specific processing configuration. The assessment does not consider arbitrary compromise of hospital infrastructure, theft of cryptographic keys, or collusion among trusted components. These limitations establish the boundaries within which privacy properties are evaluated. Operational failures involving privacy accounting and policy enforcement are considered separately as system requirements.

4 Privacy-Preserving Analytical Framework

This section presents the architectural components, information flows, and privacy-enforcement procedures through which the selected PETs are integrated into the proposed framework.

4.1 Privacy-Preserving Statistical Analytics

The statistical analytics component provides a controlled environment for executing predefined queries over distributed healthcare records. For the experimental implementation, seven representative statistical queries (`Q001-Q007`) were configured and approved to demonstrate the framework's query processing and privacy-governance capabilities. These queries illustrate the supported analytical operations rather than defining an inherent limit on the framework. Additional queries would require appropriate specification and validation of their contribution bounds, sensitivity, privacy mechanisms, and budget requirements before authorisation. Query execution is coordinated across five simulated hospital nodes, with DP applied to the combined result. Figure 1 illustrates the statistical query processing workflow.

4.1.1 Differentially Private Query Processing

The framework employs OpenDP [33] to construct privacy-preserving statistical operations under a patient-level privacy model. To bound individual contribution, the analytical dataset is restricted to a single admission per patient.

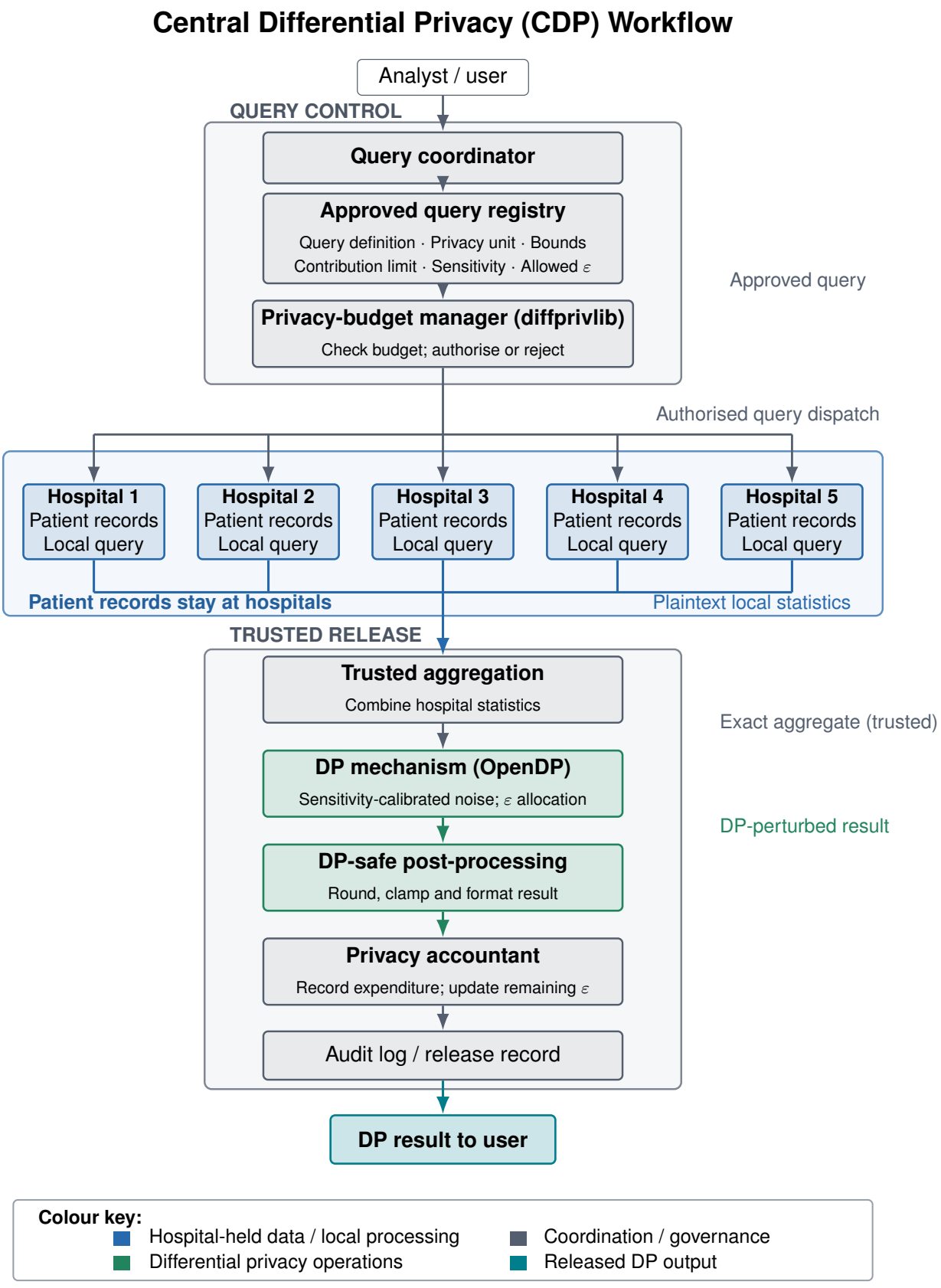


Figure 1: Central Differential Privacy (CDP) statistical query processing workflow. Colours distinguish hospital-held data, coordination and governance, differential privacy operations, and released results.

Query sensitivity is determined by the relevant aggregate operation, with explicit values bounded where required. The implemented mechanism use pure differential privacy ($\delta = 0$) with configurable privacy parameters ($\varepsilon \in \{0.1, 0.5, 1.0, 5.0, 10.0\}$). Noise is calibrated according to the sensitivity and privacy budget assigned to each query. For the mean calculation, the allocated budget is distributed between the bounded-sum and count operations used to construct the final estimate.

Following aggregation, the DP mechanism generates the protected statistical result for release. The use of predefined queries and contribution bounds ensures that the required privacy parameters and sensitivity assumptions are established before execution.

4.1.2 Privacy-Budget Management and Query Governance

The framework incorporates a privacy-governance layer to regulate repeated analytical requests. Each query is associated with a specified privacy cost, which is evaluated against the remaining budget before execution. Requests that exceed the configured limit are rejected without consuming additional privacy expenditure.

Additional controls govern high-cost operations. Queries exceeding the configured expenditure threshold require explicit confirmation through the `confirm_high_spend` parameter before processing is authorised. The system maintains a privacy ledger to record expenditure and relevant execution decisions. Successful privacy-consuming release updates the cumulative budget, whereas validation and rejected requests incur no expenditure. Together, these components establish a control query-execution procedure in which mechanism configuration, privacy budget, authorisation, statistical computation and release accounting are coordinated within a single analytical flow.

4.2 Homomorphic Encryption-Enable Analytics

The framework incorporates HE into the statistical analytical pathway to enable aggregation without exposing plaintext contributions to the analytical coordinator. The implementation uses TenSeal [34], employing the BFV scheme for integer-valued statistics, including counts and histograms, and CKKS for bounded real-valued sums. Queries that require both operations, such as the mean, employ a combination of these schemes. Figure 2 illustrates the workflow for the HE-enabled differentially private query processing.

4.2.1 Encrypted Aggregation

Each participating hospital computes the required local statistics and encrypts them using the public-key context. The encrypted contributions are transmitted to the coordinator, which performs homomorphic aggregation without accessing their plaintext values or processing the private decryption key. The resulting ciphertext represents statistics across participating hospitals. BFV supports exact integer arithmetic, whereas CKKS introduces numerical approximation in real-value computation. The framework, therefore, distinguishes between integer and floating-point workloads when selecting the encryption scheme.

4.2.2 Differential Privacy and Protected Release

The framework integrates DP into encrypted computation workflows by incorporating calibrated noise before decryption. A dedicated noise service generates the required DP noise according to the query sensitivity and allocated ε, and encrypts it using the corresponding public-key context. The encrypted noise is then combined homomorphically with the aggregated ciphertext.

The resulting ciphertext is a perturbed aggregate that is forwarded to the trusted release service for decryption. This design prevents the analytical coordinator from accessing plaintext intermediary values and avoids exposing the exact global aggregate during the intended release procedure. The validity of the intended DP guarantee depends on correct noise calibration and the numerical representation supported by the encryption scheme. Detailed implementation and experimental validation are presented in the subsequent sections.

4.3 Differentially Private Federated Learning

The federated learning component is implemented using Flower [35] and PyTorch [36] with five simulated hospital nodes participating in collaborative model training. The framework employs a logistic regression model to predict prolonged ICU length of stay using MIMIC-IV data. Model updates are aggregated using Federated (FedAvg), while differential privacy is incorporated into local optimisation. The overall federated learning procedure, including local differentially private optimisation, model aggregation, and cumulative privacy accounting, is illustrated in Figure 3.

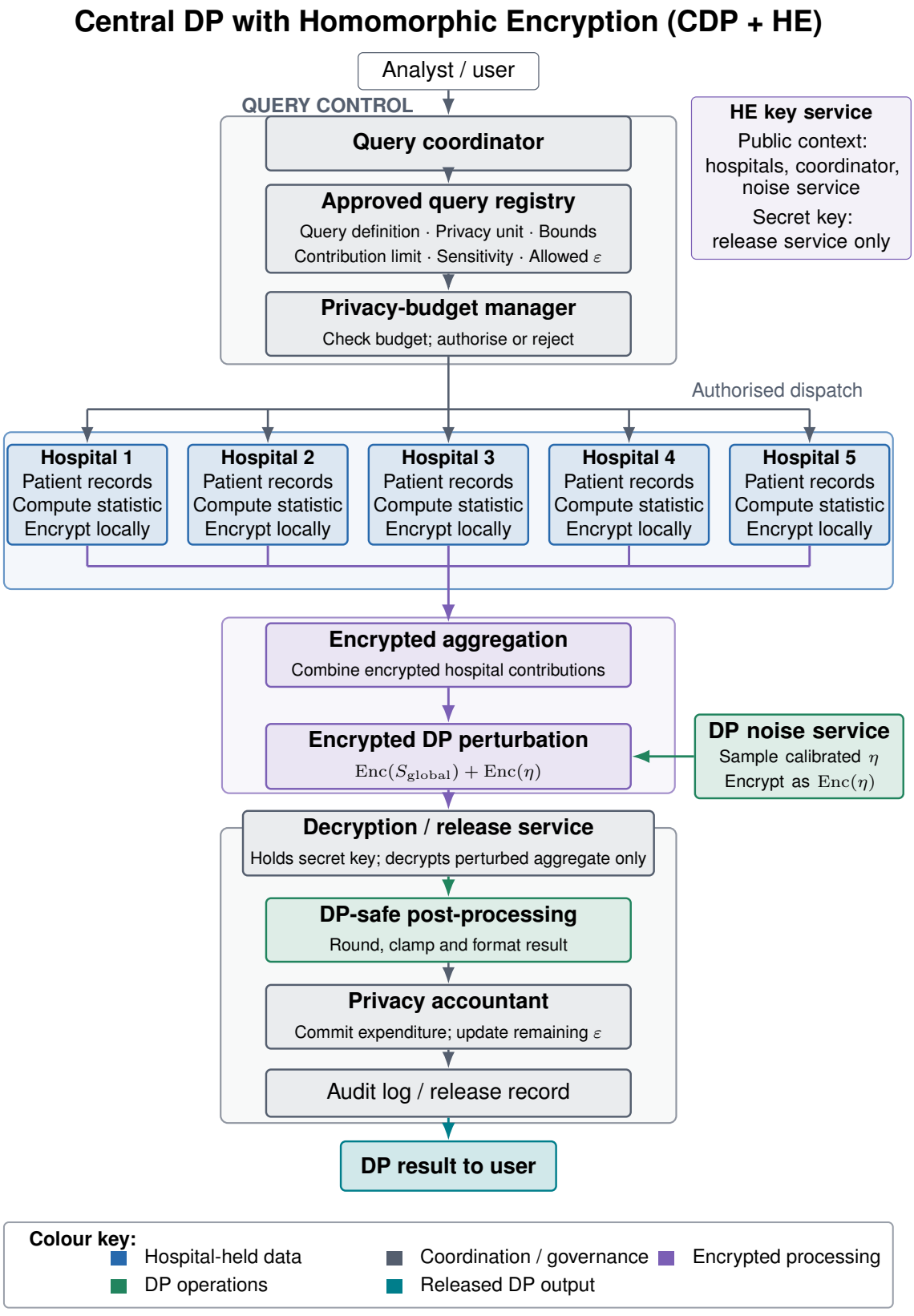


Figure 2: Homomorphic Encryption (HE)-enabled statistical analytics workflow. Colours distinguish hospital-held data, coordination and governance, encrypted computation, differential privacy operations, and released results.

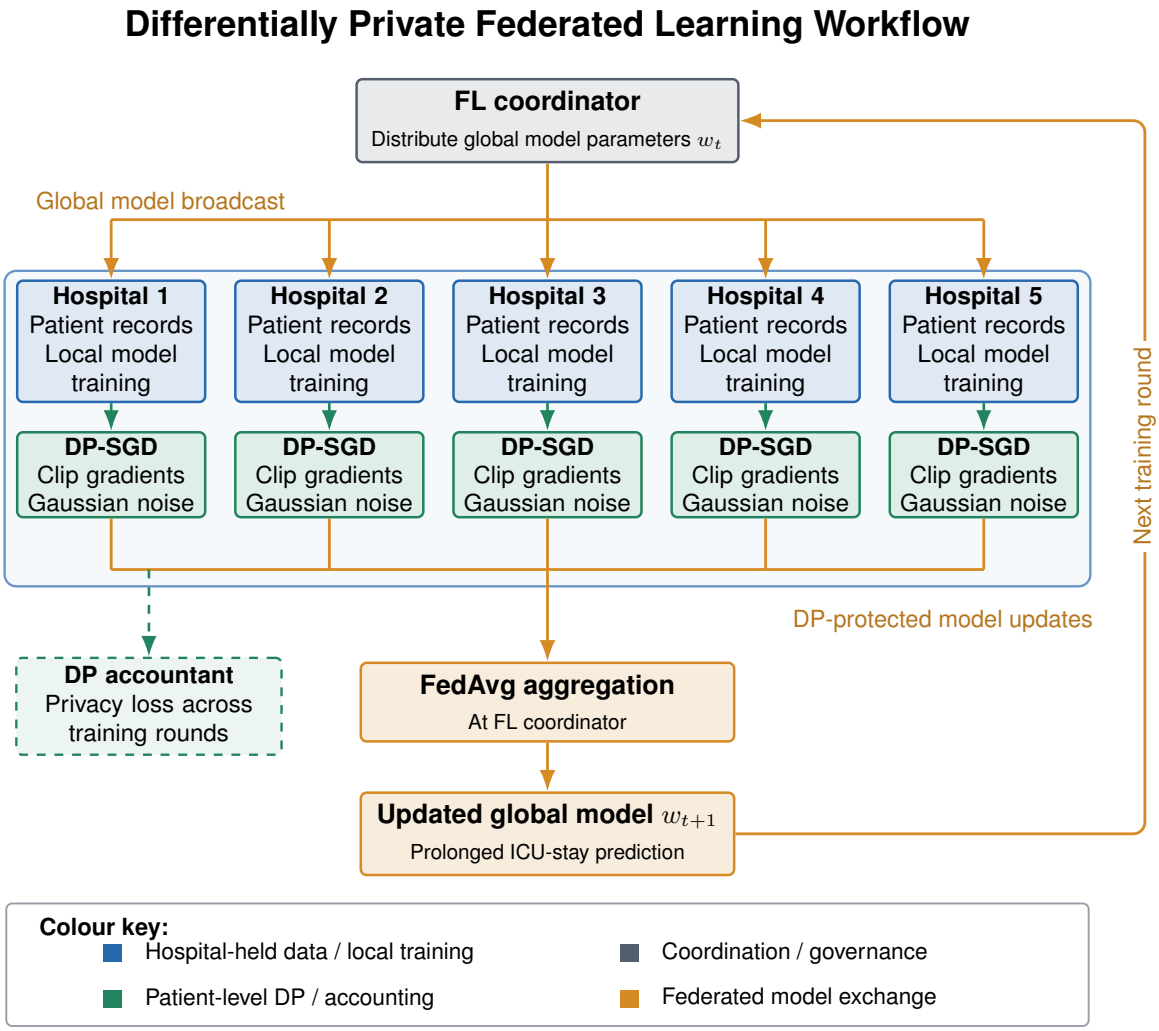


Figure 3: Differentially private federated learning workflow. Colours distinguish hospital-held data, coordination and governance, differential privacy operations, and federated model exchange.

4.3.1 Federated Training and Patient-Level Contributions

The framework considers two patient-level data representations to account for patients with multiple ICU admissions. The first retains a single eligible stay per patient, while the second incorporates up to five stays by combining their contributions into a patient-level gradient.

In the multi-stay configuration, gradients for the same patients are aggregated before clipping, ensuring that the contribution bound applies to the patient rather than to a single hospital admission. This distinction is necessary to align the privacy mechanism with the intended patient-level protection.

4.3.2 Differential Privacy and Privacy Accounting

DP is incorporated into local model optimisation through gradient clipping and the addition of Gaussian noise. The bounded patient-level gradients are aggregated and perturbed before the resulting local model update is used in federated aggregation. Privacy is evaluated using the sampling probability, clipping, noise multiplier, and number of optimisation steps. The experimental design considers hospital-specific sampling rates and noise calibration to evaluate a common privacy target across participating nodes.

The framework examines privacy-utility behaviours under different privacy budgets and contribution bounds, with cumulative accounting applied across training rounds. The corresponding implementation parameters and experimental results are presented in the subsequent chapters.

4.4 Context-Dependent Privacy Threshold and Parameter Selection

Although differential privacy provides a mathematical quantification of privacy loss, determining an acceptable privacy threshold remains a context-dependent decision. There is no universally applicable ε value that establishes sufficient protection across all healthcare applications. Appropriate privacy parameters depend on factors such as data sensitivity, the protected population, the analytical purpose, acceptable risk disclosure, and utility requirements. This challenge is also highlighted by NIST Guidelines for Evaluating Differential Privacy Guarantees, which identifies privacy-parameter selection as an area requiring careful contextual assessment [37].

The proposed framework demonstrates how different privacy and utility configurations can be achieved and evaluated, rather than prescribing a fixed privacy threshold. Through configurable parameters, expenditure limits, and experimental comparison, it enables the examination of privacy-utility trade-offs across healthcare analytical tasks. The framework, therefore, supports informed selection of privacy parameters while leaving the determination of acceptable privacy levels to the relevant organisation and application context.

5 System Implementation

This chapter presents the implementation of the proposed privacy-preserving healthcare analytics framework. It describes the experimental environment, healthcare data preparation, and software components used to implement the statistical and federated learning workflows.

5.0.1 Implementation Environment

The framework was developed in Python 3.11.4 on Windows 11 Platform equipped with a 12th-gen Intel Core i7-1260 processor and an NVIDIA T550 GPU with 4 GB of dedicated memory. Docker 29.7.2 and Docker Compose 5.5.0 were used to simulate five hospital nodes as independent containerised services that communicate with the statistical analytical coordinator via REST APIs developed with FastAPI and Uvicorn.

OpenDP 0.15.1 [33] was used for differential with IBM's diffprivlib accountant [38] for budget management. while TenSEAL 0.3..17 supported homomorphic encryption [34]. The federated learning component was developed using Flower 1.39.0 [35], PyTorch 2.114.0 [36], and Opacus 1.6.0 [39] for distributed model training and differential privacy optimisation. Supporting libraries included NumPy, Pandas, Scikit-learn [40], and SciPy.

5.1 Data Preprocessing

The framework was evaluated using the MIMIC-IV dataset, which contains de-identified patient demographics, hospital admissions, intensive care records and associated clinical information. Separate data preparation procedures were applied to support the statistical analytics and federated learning experiments.

5.1.1 Statistical Analytics Dataset

For the Statistical analytical workflow, the dataset comprised 223,452 unique patients and 546,028 hospital admissions. Records were partitioned across five simulated hospital nodes using patient identifiers `'subject_id'`, with allocation of 20%, for $\texttt{hospital}_1$, $\texttt{hospital}_2$, $\texttt{hospital}_3$, 5% to $\texttt{hospital}_4$ and 35% to $\texttt{hospital}_5$. This ensures that the records belonging to the same patient are assigned to a single node.

For privacy-preserving statistical queries, patient-level contribution limits were enforced by selecting a single eligible admission per patient when required. Numerical attributes were bounded according to their corresponding query definition, including hospital length of stay restricted to $[0, 30]$ days. The pre-processing constraint established the contribution limits and sensitivity assumptions used by the DP mechanisms.

5.1.2 Federated Learning Dataset

The federated learning workflow used a processed ICU cohort comprising 74,501 stays from 54,351 patients. The prediction task was formulated as a binary classification problem to identify prolonged ICU Length of Stay (LOS) of at least 4 days.

Feature reduction was performed before federated training to address the high dimensionality and sparsity of the original clinical representation. The initial 5,222 medication indicators were consolidated into 192 categories using the Anatomical Therapeutic Chemical (ATC) classification at the third level (ATC3) [41]. Laboratory features were reduced from 458 to 120 by retaining measurements with at least 5% observation coverage before imputation. Following these

reductions, two constant medication indicators were removed, producing a final set of 351 predictors, comprising 193 binary and 158 continuous features.

This reduction was intended to improve computational efficiency and limit the influence of highly sparse features during differentially private optimisation, while retaining clinically relevant information. Binary variables were preserved in their original representation, whereas continuous variables underwent a bounded signed-log transformation. The resulting data was checked for missing and non-finite values.

Patients were distributed across five simulated hospitals using the same allocation proportions as before, ensuring patient-disjoint partitions. within each hospital, and an 80:20 patient-level training and test split was applied, resulting in 43,450 training patients and 10,864 testing patients overall. The prepared dataset supported by both single-stay and multiple-stay representations for the federated learning experiments.

5.2 Differential Privacy Query Implementation

The implementation supports seven predefined queries, each associated with the relevant statistical operation, contribution, constraints, sensitivity, and privacy-budget allocation.

5.2.1 Query Implementation

The predefined queries were implemented with their corresponding statistical operations, contribution constraints and sensitivity configurations. OpenDP was used for noise generation and diffprivlib, accounting for budget management. Whereas the privacy expenditure was managed through a reservation-and-commit procedure, with execution and governance information recorded in persistent JSONL audit files. The implementation incorporated request validation, high-spend confirmation and synchronisation controls for concurrent operations to maintain consistency between execution and privacy-budget records. These controls were examined through functional tests covering successful releases, rejected requests, and budget-enforcement behaviour.

5.2.2 Homomorphic Encryption Implementation

The HE components were implemented using TenSEAl, the BFV scheme was used for integer-based operation (`Q001,Q002, Q004, and Q006`), while queries involving bounded numerical aggregates `Q003,Q005, and Q007` employ CKKS for real-valued sums and BFV for associated counts.

The implementation separates cryptographic operations across dedicated services. Hospital nodes generate encrypted contributions, which are serialised and transmitted for homomorphic aggregation. Public encryption contexts are available to the coordinator and noise service, while the corresponding private decryption contexts remain within the release service. For the encrypted noise implementation, the noise service generates query-specific DP noise using integer-valued Laplace noise for BFV operations and continuous noise for CKKS based sums. The noise is encrypted and combined with aggregated ciphertexts before decryption.

The implementation was evaluated for the correctness of encrypted computation, numerical fidelity, and privacy-preserving release behaviour. The corresponding validation procedures and performance measurements are presented later. The implemented statistical workload comprises seven predefined queries covering population counts, categorical distributions, bounded means, and rare-subgroup analysis. Table 3 summarises their respective operations, sufficient statistics, and privacy mechanisms. Whereas Figure 4 represents the CDP query execution path.

5.3 Federated Learning Implementation

The FL component was implemented using Flower 1.39.0 and PyTorch 2.14.0 with Opacus 1.60 supporting differential privacy optimisation. Five simulated hospital clients participated in training a logistic regression model to predict ICU length of stay. The Flower server coordinated training rounds and aggregated locally trained model parameters using Federated Averaging (FedAvg).

The implementation supports two patient-level training configurations: a single ICU stay per patient and multiple stays, with up to five stays contributing to an individual patient's grade. In the multi-stay configuration, for an individual patient's gradient. In this multi-stay configuration, gradients associated with the same patient are combined before clipping to preserve the intended patient-level contribution bound. Gaussian noise is subsequently applied during local optimisation.

Privacy accounting incorporates the sampling configuration, noise multiplier, and cumulative optimisation steps. Hospital-specific sampling rates and noise configurations enable evaluation of the training procedure under defined privacy targets. The corresponding experimental settings and comparisons are presented in the evaluation chapters.

Table 3: Statistical queries implemented in the healthcare analytics framework.

ID	Statistical query	Local sufficient statistic	DP mechanism
Q001	Unique patient count	Scalar patient count	Laplace mechanism; sensitivity 1.
Q002	Admission-type histogram	9-bin count vector	Vector Laplace mechanism; ℓ_1 sensitivity 1.
Q003	Bounded mean length of stay	Bounded LOS sum and patient count	Separate noisy sum and count; ϵ split equally.
Q004	Race $\times$ admission-type distribution	33×9 joint histogram (297 cells)	Vector Laplace mechanism; ℓ_1 sensitivity 1.
Q005	Mean length of stay by race	33 grouped bounded sums and 33 counts	Separate noisy sum and count vectors; ε split equally.
Q006	Rare-subgroup patient count (Asian–Korean, urgent admission)	Subgroup patient count	Laplace mechanism; sensitivity 1.
Q007	Rare-subgroup mean length of stay	Subgroup-bounded LOS sum and patient count	Separate noisy sum and count; ε split equally.

Note: Each patient contributes through their earliest eligible admission. LOS values are bounded to $[0, 30]$ days. Mean queries use DP-safe post-processing.



Figure 4: Execution sequence of a differentially private statistical query.

6 Evaluation of Differentially Private Statistical Analytics

This chapter investigates whether Central Differential Privacy (CDP) can support useful and controlled statistical analytics under conditions relevant to distributed healthcare environments. Rather than considering the statistical accuracy of DP mechanisms in isolation, the experiments examine a sequence of questions concerning analytical utility, incomplete data availability, information disclosure, and operational privacy governance.

The evaluation first establishes how query characteristics, population size, and privacy-budget selection influence statistical accuracy. It then examines whether unequal patient distributions and missing hospital contributions affect the reliability of the released statistics. Controlled disclosure experiments subsequently investigate what an adversary may infer from aggregate results under different knowledge assumptions. Finally, the evaluation considers controlling query authorisation and budget enforcement so that cumulative privacy expenditure can be governed.

Together, these experiments assess the conditions under which Privacy-preserving statistical releases remain informative, the risks associated with their use, and the operational controls required to support repeated analytical requests.

6.1 When Do Differentially Private Statistics Remain Analytically Useful?

The statistical utility evaluation examined the effect of population size on differentially private counts and bounded means. For each query and privacy configuration, 1000 independent Monte Carlo releases were generated from fixed statistics. Utility was assessed using mean absolute error (MAE) and mean relative absolute error.

Population Counts and Subgroup Rarity. Queries `Q001` and `Q006` were compared to investigate the effect of subgroup prevalence under equivalent privacy mechanisms. The exact counts were 223,452 patients for `Q001` and 38 patients for `Q006`.

The result should have a comparable absolute error but a substantially different relative error. At $\varepsilon = 1$, `Q001` produced an MAE of 0.980 patients, compared with 0.954 for `Q006`. However, the mean relative error increased from 0.00044% for `Q001` to 2.51% for `Q006`. At $\varepsilon = 0.1$ the corresponding relative errors were 0.00425% and 25.53%

The difference arose because an equivalent absolute perturbation represents a substantially larger proportion of a small count. Furthermore, at $\varepsilon = 0.1, 1, 3\%$ of `Q006` releases were post-processed to zero, demonstrating the influence of non-negative clipping on sparse statistical outputs.

Bounded Mean Estimates. The second experiment compared `Q003` and `Q007`, which estimate bounded hospital length of stay for the computation and a rare subgroup, respectively. Their exact mean values were 4,266 and 4.590 days, while their supporting populations comprised 223,452 38 patients.

At $\varepsilon = 11$, `Q003` achieved an MAE of approximately 0.000279 days, whereas `Q007` produced an MAE of 1.515 days. The difference became particularly pronounced at $\varepsilon = 0.1$ where `Q007` recorded an MAE of 9.340 days. At this privacy setting, 35.4% of its releases reached the lower bound of zero days and 16.3% reached the upper bound of 30 days.

Unlike scalar counts, bounded means are perturbed by both their numerator and denominator. The small supporting population in `Q007` of the variability of the noisy ratio, while domain clipping introduces additional distribution effects.

Table 4: Statistical utility of common and rare-population queries across privacy budgets.

	Q001	Q006	Q003	Q007
ϵ	Rel. error (%)	Rel. error (%)	MAE (days)	MAE (days)
0.1	0.00425	25.53	0.002736	9.340
0.5	0.000846	5.38	0.000563	2.833
1.0	0.000439	2.51	0.000279	1.515
5.0	0.0000389	0.237	0.0000527	0.305
10.0	0.00000358	0.0211	0.0000267	0.158

Table 4 summarises the matched-query comparisons across the evaluated privacy parameters. The findings demonstrate that the practical utility of a DP release in a distributed environment depends on the supporting population and statistical operation, rather than on ε alone.

Evaluation of Alternative Mean-Release Mechanisms. The bounded mean query `Q003` was evaluated using two alternative release mechanisms: the implemented noisy sum/count construction and direct perturbation of the mean. The comparison retained patient-level add/remove adjacency and examined the mean absolute error over 1,000 Monte Carlo release per privacy setting.

Under add/remove adjacency, the direct mean has global sensitivity if 20 days for the bounded domain $[0, 30]$, where the implemented construction separately perturbs the bounded sum and patient count, allocating half the privacy to each operation.

The results in Table 5 show that the sum/count mechanism consistently achieved substantially lower error. At $\varepsilon = 1$ its MAE was 0.000283 days, compared with 8.152 days for direct mean perturbation. This remains substantial across all evaluated privacy parameters.

A secondary experiment considers a direct mean under replace-one adjacency with a fixed population size. At $\varepsilon = 1$, this alternative achieved an MAE of 0.000131 days, compared with 0.000278 days for the sum/count mechanism in that experiment. However, the comparison changes the neighbouring-dataset definition and therefore does not establish superiority under an equivalent privacy model.

Table 5: Mean absolute error of Q003 release mechanisms under patient-level add/remove adjacency.

ϵ	Sum/count MAE (days)	Direct mean MAE (days)
0.1	0.002592	13.851
0.5	0.000537	10.995
1.0	0.000283	8.152
5.0	0.0000521	2.587
10.0	0.0000256	1.439

The findings support the implemented sum/count construction for the adopted add/remove-patient privacy model for this configuration. They also demonstrate that comparisons between DP mechanisms must account for their adjacency assumption, since changes to the privacy definition can substantially affect the required noise calibration. Therefore, the same privacy budget is not equivalent under two different adjacency models.

6.2 How Do Hospital Heterogeneity and Missing Data Affect DP Results?

A central concern in distributed healthcare analytics is whether Privacy-preserving computation remains reliable when participating hospitals differ in patient composition or cannot provide complete data. This experiment, therefore, examines whether observed errors arise from differential privacy itself or from changes in the population represented by the available data. Two scenarios were considered using the rare-subgroup queries `Q006` and `Q007`. The first redistributes subgroup patients across hospitals while preserving the complete analytical population. The second progressively removes selected subgroup contributions, representing incomplete data availability. By distinguishing changes in the underlying statistical target from DP-induced perturbation, the experiments assess whether missing data compound the utility challenges associated with privacy-preserving statistical releases.

Site Heterogeneity. The redistribution experiment progressively reassigned the 14 target-subgroup patients originally held by Hospital 5 to the remaining hospitals. Although the local subgroup distributions changed, the global count (`Q006`) remained 38 patients, and the bounded mean length of stay (`Q007`) remained 4.590 days.

At $\varepsilon = 1$, the mean absolute error (MAE) ranged from 0.949 to 0.972 patients for `Q006`, and from 1.525 to 1.601 days for `Q007`. These variations are consistent with Monte Carlo variability rather than changes in the underlying statistical target.

From a distributed analytics perspective, the results empirically confirm *partition invariance*: unequal subgroup representation across participating hospitals does not alter the global statistical target or the theoretical central-DP output distribution when all patient contributions are included. This property follows from patient-disjoint partitioning, correct aggregation of local sufficient statistics, and applying DP once to the combined result.

Data Unavailability. In the second scenario, selected subgroup patients were removed rather than redistributed, changing the population represented by the aggregate. When all 14 target-subgroup patients originally held by Hospital 5 were removed, the `Q006` count decreased from 38 to 24, representing a 36.84% reduction. The exact `Q007` mean decreased from 4.590 to 3.831 days.

To distinguish the effect of incomplete data availability from DP perturbation, the signed total release error was decomposed as

$$\underbrace{\widetilde{\theta} - \theta_{\text{full}}}_{E_{\text{total}}} = \underbrace{\theta_{\text{available}} - \theta_{\text{full}}}_{E_{\text{availability}}} + \underbrace{\widetilde{\theta} - \theta_{\text{available}}}_{E_{\text{DP}}}, \tag{2}$$

where θ_{full} denotes the complete-population statistic, $\theta_{\text{available}}$ the statistic computed from available observations, and $\widetilde{\theta}$ the corresponding DP release.

At $\varepsilon = 1$, the `Q006` DP-only MAE remained approximately 0.98 patients following complete removal of Hospital 5's target subgroup. However, its total MAE relative to the complete-population reference increased to 13.978 patients. For `Q007`, DP-only MAE increased from 1.574 to 2.396 days as the supporting population decreased from 38 to 24 patients. Table 6 summarises the observed effects.

Table 6: Effect of subgroup data unavailability on distributed DP statistics at $\varepsilon = 1$. Total MAE is measured relative to the complete-population reference.

H5 subgroup retained (%)	Q006 exact count	Q007 exact mean (days)	Q006 total MAE (patients)	Q007 total MAE (days)
100	38	4.590	0.983	1.574
75	35	4.638	3.051	1.704
50	31	3.717	6.975	1.931
25	28	3.849	9.923	2.075
0	24	3.831	13.978	2.484

The experiments distinguish two conditions relevant to distributed healthcare analytics. Heterogeneous patient allocation preserves statistical correctness when the complete population contributes to aggregation. In contrast, incomplete availability changes the underlying statistical target and can amplify DP-induced error for small supporting populations. *Consequently, the reliability of distributed DP outputs depends not only on noise calibration but also on the completeness of participating data sources.* These controlled experiments assess subgroup-level data loss.

6.3 What Can an Adversary Infer from DP-protected Statistical Releases?

To complement the statistical utility evaluation, controlled disclosure experiments examined whether aggregate releases could reveal a patient's prolonged hospital-stay status. Three experimental configurations were considered: attribute inference from subgroup statistics (`Q008`), inference under varying auxiliary knowledge (`Q008-B`), and overlapping-query differencing (`Q009`). These were conducted as separate experimental benchmarks rather than unrestricted queries within the operational platform.

The experiments were motivated by the actual MIMIC-IV subgroup distribution, where 77 non-empty race–admission-type groups contained 20 or fewer patients, including two singleton groups. Controlled synthetic cohorts were subsequently used to isolate individual contributions and evaluate attack behaviour under reproducible conditions.

Attribute Inference and Auxiliary Knowledge. The `Q008` experiment considered a target patient's prolonged-stay status, defined by a hospital length of stay exceeding 14 days. Two otherwise identical datasets represented a non-prolonged (7-day) and prolonged (20-day) target admission. The attacker observed a released two-cell histogram and knew the prolonged-stay statuses of all other cohort members.

Cohort sizes of $n \in \{5, 10, 20, 50, 100\}$ were evaluated across five privacy budgets, with 1,000 independent repetitions per target state. Exact releases enabled perfect inference under the specified auxiliary-knowledge assumption. In contrast, mean attack accuracy under DP ranged from 52.67% at $\varepsilon = 0.1$ to 99.63% at $\varepsilon = 10$, as reported in Table 7.

The `Q008-B` experiment extended this analysis by varying the proportion of known non-target patient statuses from 0% to 100% in cohorts of 5, 10, and 20 patients. Under complete auxiliary knowledge, exact-release attack accuracy was 100%. At $\varepsilon = 0.1$, DP reduced the observed attack accuracy to 52.85%, 52.58%, and 53.83%, respectively. These results show that strong perturbation can substantially reduce the additional information available from a release, even when the adversary possesses extensive background knowledge.

However, utility deteriorated for smaller cohorts. At $\varepsilon = 1$, the mean absolute error of the released prolonged-stay rate was 15.36 percentage points for $n = 5$, compared with 0.80 percentage points for $n = 100$. Thus, increasing cohort size improved statistical utility but did not necessarily reduce inference success when the attacker could eliminate the known background contributions.

Figure 5 quantifies the utility limitations of differentially private subgroup statistics across cohort sizes and privacy budgets. Considered alongside the attribute-inference results, the findings highlight that improved statistical accuracy in larger cohorts does not necessarily imply reduced disclosure risk under extensive auxiliary knowledge. This distinction is relevant when evaluating the suitability of privacy configurations for healthcare subgroup analysis.

Figure 6 illustrates the effect of increasing auxiliary knowledge on attribute-inference accuracy for a five-patient cohort under different privacy budgets. The horizontal baseline at 0.5 represents chance-level target inference under the balanced experimental design. Attack accuracy approaching this baseline indicates that the released statistic provides limited additional discriminatory information to the evaluated adversary.

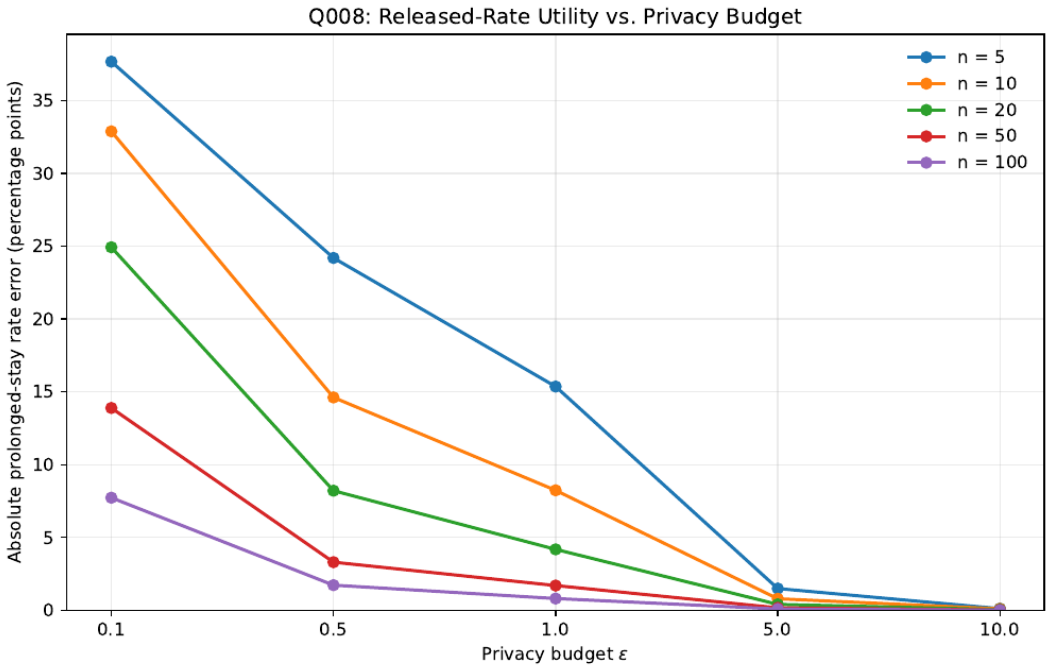


Figure 5: Mean absolute error of the differentially private prolonged-stay rate (Q008) across privacy budgets and cohort sizes, expressed in percentage points.

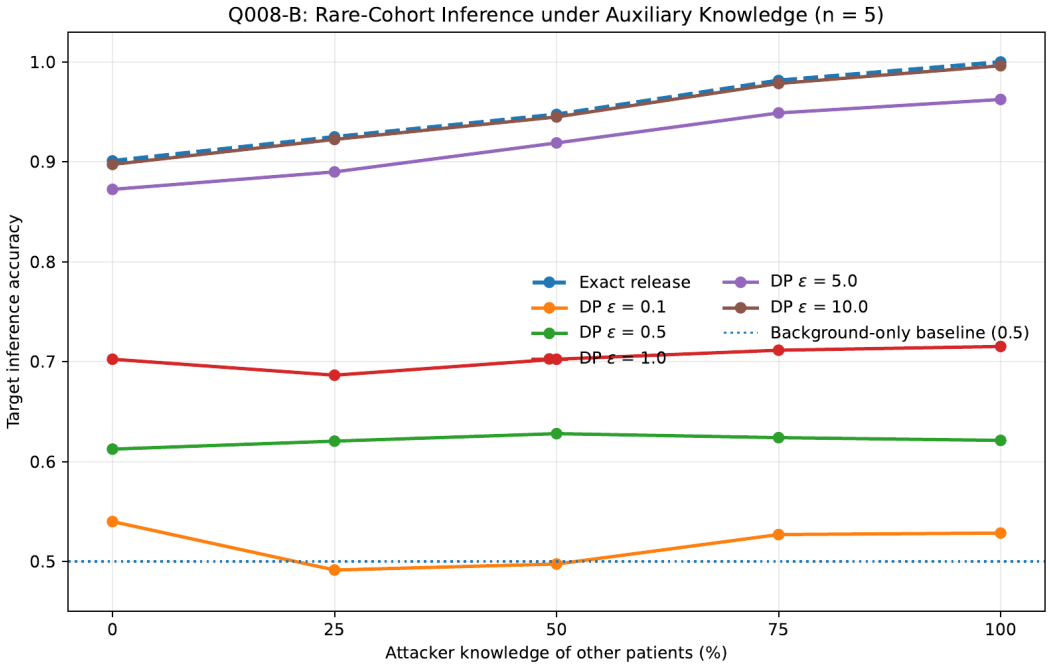


Figure 6: Effect of auxiliary knowledge on attribute-inference attack accuracy for a five-patient cohort (Q008-B) across different privacy budgets (ε).

Overlapping-Query Differencing. The Q009 experiment evaluated whether overlapping aggregate queries could isolate an individual's contribution. Two counts were considered:

$$Q_1 = Q(D), \qquad Q_2 = Q(D \setminus \{t\}), \tag{3}$$

where t denotes the target patient. Their exact difference reveals whether the target contributes to the prolonged-stay count.

For the DP experiment, independent Laplace noise was applied to both outputs, with the total privacy budget divided equally between the two releases:

$$\varepsilon_{\text{total}} = \varepsilon_{Q_1} + \varepsilon_{Q_2}, \qquad \varepsilon_{Q_1} = \varepsilon_{Q_2} = \frac{\varepsilon_{\text{total}}}{2}. \tag{4}$$

Across the evaluated cohort sizes, exact differencing recovered the target contribution with 100% accuracy. Under DP, mean attack accuracy was 50.12% at $\varepsilon_{\text{total}} = 0.1$, increasing to 90.51% at $\varepsilon_{\text{total}} = 10$. At a total privacy budget of $\varepsilon = 1$, the differencing attack achieved 56.51% accuracy, with a mean absolute reconstruction error of 2.989 patients.

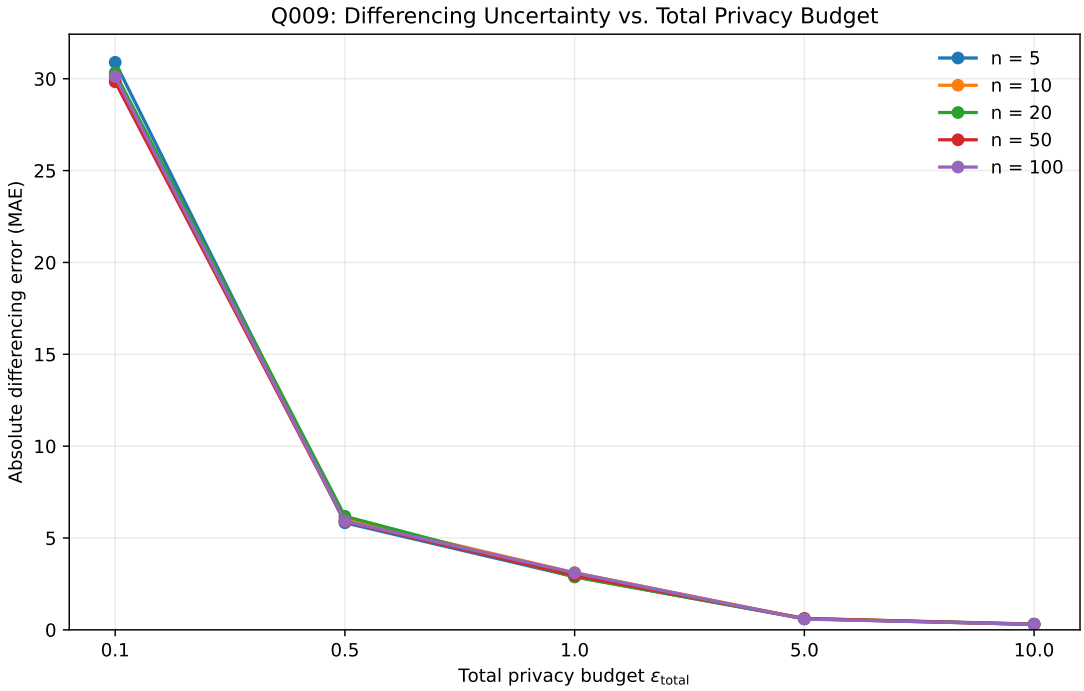


Figure 7: Mean absolute reconstruction error under overlapping-query differencing (Q009) across total privacy budgets and cohort sizes. The privacy budget is divided equally between the two differentially private releases.

Figure 7 illustrates the reconstruction error under overlapping-query differencing. The similar error across cohort sizes indicates that, under the evaluated attack configuration, increasing the number of patients does not substantially increase uncertainty about the isolated contribution.

Interpretation and Limitations. The experiments demonstrate that aggregate size alone is insufficient to characterise disclosure risk. Auxiliary knowledge can isolate an individual's contribution, while overlapping releases introduce additional opportunities for inference. The Q009 results further illustrate the importance of accounting for cumulative privacy expenditure across related queries.

These findings are conditional on the specified attacker models and controlled datasets. In particular, the Q008 target-state comparison represents a two-step replacement under the platform's add/remove-patient adjacency, while Q009 is an experimental differencing benchmark rather than an authorised operational query. The measured attack accuracies, therefore, complement, but do not establish, the formal DP guarantee. The experiments assess disclosure through released outputs rather than attacks against hospital infrastructure or distributed communication channels.

Table 7: Controlled disclosure-attack performance across privacy budgets. Values are averaged over five cohort sizes.

ε	Q008 Accuracy	Q008 AUC	Q009 Accuracy	Q009 AUC
0.1	0.5267	0.5302	0.5012	0.5016
0.5	0.6146	0.6448	0.5397	0.5482
1.0	0.7104	0.7667	0.5651	0.5802
5.0	0.9633	0.9958	0.7683	0.8244
10.0	0.9963	1.0000	0.9051	0.9593

6.4 Can Repeated Statistical Queries be Governed within a Privacy Budget

The governance evaluation examined whether the framework could enforce cumulative privacy constraints while maintaining accountability for statistical releases across participating hospitals. Since repeated DP queries consume a shared privacy budget, controlling individual requests is insufficient without considering their cumulative expenditure. The framework therefore enforces a configurable privacy budget ceiling, under which each requested ε is evaluated against the remaining allocation before distributed computation is authorised.

Privacy-Budget constraints. A request may satisfy the configured budget limit while consuming a substantial proportion of the available privacy allocation. To reduce unintended expenditure, the framework incorporates a high-spend confirmation requirement when

$$\frac{\varepsilon_{\text{request}}}{\varepsilon_{\text{total}}} \geq 0.25 \quad \vee \quad \frac{\varepsilon_{\text{request}}}{\varepsilon_{\text{remaining}}} \geq 0.50. \tag{5}$$

Requests meeting either condition require explicit authorisation through `confirm_high_spend=true`. Without confirmation, the request is clocked before hospital dispatch, and no privacy expenditure is committed. Requests exceeding the remaining budget are rejected regardless of confirmation.

These thresholds represent configurable operational safeguards rather than formal privacy risk boundaries. They do not alter the DP mechanism or its mathematical guarantee; instead, they provide an additional decision point before consuming a substantial proportion of the available budget.

Privacy Accounting and Auditability. The framework maintains separate records for privacy expenditures and execution auditing. The privacy ledger records committed expenditure and supports the calculation of cumulative and remaining privacy budgets. Expenditure is checked before query execution and committed following a successful DP release. The audit trail records request identifiers, query definitions, privacy parameters, execution status, the count of participating hospitals, and relevant error information. This enables authorised operations, policy-based rejections, and distributed-execution failures to be distinguished without storing patient-level records or exact intermediate statistics in the audit log.

This separation supports operational accountability while limiting unnecessary exposure of sensitive analytical information. Importantly, auditing in the proposed framework provides traceability of system decisions but does not independently establish compliance with the formal DP guarantee.

6.5 Discussion and Lessons Learned: What do these Experiments Establish about the Suitability of CDP?

The valuation highlights a central challenge in the practical adoption of Differential privacy: although privacy loss can be mathematically quantified through ε, determining an acceptable privacy threshold remains a context-dependent decision. A particular value of ε does not independently establish whether the resulting protection is sufficient for a healthcare application. Such a judgment requires consideration of the protected population, the sensitivity of the underlying information, the intended analytical purpose, the potential consequences of disclosure, and the acceptable loss of utility.

The experimental framework provides a practical means of examining these considerations rather than prescribing a universally appropriate privacy budget or configuration. By varying parameters across healthcare queries, the evaluation demonstrates that the same privacy configuration can yield substantially different analytical outcomes. Population-level statistics may retain useful accuracy when rare-subgroup estimates become reliable. A similar privacy budget yields different statistical utility across different adjacencies. Similarly, the disclosure experiments show that improving

statistical utility does not necessarily imply a corresponding reduction in inference risk, particularly when an adversary possesses auxiliary information. This indicates that the selection of privacy parameters should be evaluated against both analytical requirements and the disclosure risk associated with a particular use case.

The distributed experiments introduce an additional consideration for healthcare organisations operating across multiple data holders. While unequal patient distributions across hospitals did not affect the global statistical target under complete participation, missing observations introduce error independently of the DP mechanism. This distinction demonstrates that the reliability of privacy-preserving analytics depends on the underlying availability and processing conditions, rather than on privacy parameters alone. Consequently, increasing the privacy budget cannot necessarily compensate for incomplete or unrepresentative healthcare data.

While these experiments establish the utility constraints, disclosure considerations, and governance requirements of the CDP-based statistical workflow, they do not address the confidentiality of exact intermediate statistics during computation. This remaining exposure motivates the evaluation of HE as a complementary protection mechanism in the following chapter.

7 Evaluation of Homomorphic Encryption-Enabled Analytics

The preceding chapter examined the disclosure risks and utility constraints associated with differentially private statistical releases. However, protecting the released result does not necessarily prevent authorised processing components from accessing exact intermediate statistics. This raises a further architectural question: can Homomorphic Encryption (HE) reduce plaintext exposure during distributed aggregation while preserving the statistical functionality and privacy controls established in the CDP workflow?

This chapter addresses this question through a sequence of experiments examining cryptographic correctness, protected release, and computational performance. First, encrypted aggregation is compared with plaintext reference computations to establish whether the required statistical operations can be performed with acceptable numerical fidelity. The evaluation then examines whether calibrated DP noise can be incorporated into the encrypted aggregate before decryption, thereby changing the information accessible to the release service. Finally, the additional execution time and ciphertext communication requirements are measured to assess the cost of this protection.

The experiments distinguish the confidentiality provided by encrypted computation from the disclosure guarantees of DP, while considering the trust assumptions and operational dependencies retained by the combined architecture.

7.1 How Can the Additional Protection offered by HE be Evaluated?

The evaluation used the seven statistical queries `Q001-Q007` in the CDP framework. Integer-valued statistics are processed using BFV, while bounded real-valued sums employ CKKS, with BFV used for associated counts. Three computational configurations provide the basis for comparison: plaintext aggregation without perturbation, plaintext aggregation with DP, and encrypted aggregation with DP noise incorporated before decryption.

The earlier HE configuration, in which the exact aggregate is decrypted before DP perturbation, was retained solely as an experimental baseline to assess the additional cost and the change in plaintext exposure introduced by encrypted-noise processing. The exact plaintext aggregates serve as reference values for verifying homomorphic computation. Comparisons between DP-only and HE-enabled DP releases examine statistical consistency, accounting for the distinction between numerical approximation and random perturbation. Computational performance is assessed through the individual cryptographic operations and end-to-end query execution.

The evaluation aims to determine whether intermediate aggregate confidentiality can be introduced while preserving the intended statistical functionality, and to quantify the computational costs of this additional protection.

7.1.1 Which Components Require Access to Plaintext and Decryption Keys?

Key Generation and Management. The implementation employs a separate TenSEAL context for BFV and CKKS. The public context contains the corresponding HE parameters and public encryption key, enabling encryption and homomorphic evaluation without revealing the underlying plaintext. The private context additionally contains the secret key required for decryption. The implementation generates the contexts once and serialises separate public-only and private representations for subsequent service use.

The public context is distributed to participating hospital services, the analytical coordinator, and the DP noise service. Private contexts are retained within the trusted release-service boundary. The context files are stored separately in the

protected runtime directories, excluded from version control. This arrangement enables encryption and homomorphic aggregation without granting decryption capability to the coordinating or noise generation components.

Table 8 summarises the cryptographic parameters used in the experiments.

Table 8: Homomorphic encryption parameters used in the experimental implementation.

Parameter	**BFV**	**CKKS**
Polynomial modulus degree	8192	8192
Plaintext modulus	1,032,193	–
Coefficient modulus	Library default	$[60, 40, 40, 60]$ bits
Global scale	–	2^{40}

Encrypted Computation and Release. In this configuration, hospitals encrypt locally computed sufficient statistics before transmission, and the coordinator performs homomorphic aggregation using public contexts. The noise service samples query-specific DP noise, encrypts it using the corresponding public context and combines it homomorphically with the encrypted global statistic. The release service subsequently decrypts the already perturbed aggregate and performs DP-safe post-processing without introducing additional noise.

For an aggregate S and sampled noise η, the latter operation is expressed as

$$C_{\mathrm{DP}} = \mathrm{Enc}(S) + \mathrm{Enc}(\eta) = \mathrm{Enc}(S + \eta). \tag{6}$$

For bounded-mean queries, the sum and count are processed separately using CKKS and BFV, respectively, with the allocated privacy budget between the two components. This modification changes the plaintext exposure at the release boundary: the release services processes perturbed rather than exact aggregate values.

7.2 Does Encrypted Aggregation Preserve Statistical Correctness?

The correctness evaluation examined whether HE aggregation reproduced the reference statistics computed for the five hospital partitions. The validation was performed separately from DP perturbation to distinguish cryptographic approximation from randomisation error. Encrypted local contributions were aggregated and decrypted within an isolated correctness test, with the reconstructed statistics compared against their plaintext equivalents.

BFV reproduced integer-valued statistics exactly. For `Q006`, the decrypted global count was 38 patients, matching the plaintext reference without discrepancy. For `Q007`, the BFV count also remained exact, while CKKS reconstructed the bounded length-of-stay sum with an absolute error of approximately 2.04×10^{-9} days. The resulting mean differed from it reference value by approximately 5.38×10^{-11} days.

Table 9: Numerical correctness of encrypted aggregation for the rare-subgroup queries.

Query	Statistic	Plaintext	Decrypted HE	Absolute error
`Q006`	Count	38	38	0
`Q007`	Count	38	38	0
`Q007`	Bounded sum	174.4270833333	174.4270833313	2.04×10^{-9}
`Q007`	Bounded mean	4.5901864035	4.5901864035	5.38×10^{-11}

The CKKS approximation errors remained substantially below the predefined validation tolerance of 10^{-4}. This highlights that the evaluated encrypted aggregation operation preserves the numerical fidelity required for the implemented statistical queries. Consequently, the substantial error observed in the rare-subgroup DP experiments cannot be attributed to the homomorphic aggregation. The numerical errors introduced by homomorphic aggregation were negligible relative to the observed DP perturbation. The substantial utility loss in rare-subgroup queries is primarily attributable to the limited supporting population, which amplifies the effect of noise added to the bounded sum and count. The resulting instability is further influenced by domain-constrained post-processing, particularly under smaller privacy budgets.

7.3 Can DP be Incorporated Before Aggregate Decryption?

The encrypted-noise evaluation explained whether differential privacy could be incorporated before decryption, without exposing exact global aggregates to the trusted release service. The assessment considered the correctness of the encrypted perturbation, the information accessible to the individual services, and the consistency of the final release procedure.

Validation of Encrypted Perturbation. The noise-before-decryption configuration was evaluated for all seven queries using fixed privacy parameters. The implementation applied integer-valued Laplace noise under BFV for counts and histograms, while CKKS supported continuous Laplace perturbation of bounded sums. Hybrid mean queries applied noise independently to the encrypted sum and count using their respective privacy budget allocations.

The validation records confirmed that encrypted noise was completed before the release stage. The noise service reported no private key availability, and its response contained only encrypted aggregates. Following the decryption, the release service performed only the required DP-safe post-processing without additional perturbation. The validation confirmed that DP perturbation was incorporated into the encrypted aggregate before decryption, ensuring that the trusted release service processed only the perturbed statistics during the evaluated workflow. Nevertheless, the release service retains the private decryption keys, and the confidentiality guarantees remain dependent on the secure management of these keys and the enforcement of the authorised release procedure.

7.4 What is the Computational Cost of HE-Enabled Processing

The performance evaluation examined the computational and communication costs associated with incorporating differential privacy noise under encryption. End-to-end query latency was measured across seven statistical queries, five privacy budgets, and three repetitions.

End-to-End Query Performance. Table 10 presents the mean execution times across the evaluated privacy budgets. The encrypted noise configuration required an average of 1.348 s per query, compared with 1.079 s for the baseline, resulting in an additional 0.269 s, or approximately 24.9% latency.

Table 10: End-to-end execution time and encrypted aggregate size across the statistical queries.

Query	HE+CDP (s)	Encrypted noise (s)	Overhead (%)	Ciphertext (MiB)
Q001	0.614	0.809	31.8	0.412
Q002	0.919	1.114	21.2	0.412
Q003	1.297	1.639	26.4	0.729
Q004	1.066	1.309	22.8	0.412
Q005	1.380	1.724	24.9	0.729
Q006	0.924	1.150	24.5	0.412
Q007	1.350	1.688	25.0	0.729
Mean	**1.079**	**1.348**	**24.9**	–

The absolute latency increase was greatest for the hybrid (CKKS-BFV) queries (`Q003`, `Q005`, and `Q007`), ranging from approximately 0.338 to 0.344 s. These queries require separate encrypted processing of bounded sums and counts. The observed increase is consistent with their additional processing and communication requirements, although the benchmark does not isolate the execution time of each internal operation. The measured ciphertext sizes were approximately 0.412 MiB for BFV-only queries and 0.729 MiB for hybrid CKKS-BFV queries, reflecting the additional encrypted components required for bounded-mean computations. Although encrypted noise introduces additional communication between services, its measured end-to-end overhead remained approximately 24.9% across the evaluated workload.

7.5 Discussions and Lessons Learned: What additional Protection does HE provide, and What Risks Remain

The HE evaluation investigates whether the trust requirements of the original CDP could be reduced by protecting intermediate statistics throughout distributed computation without exposing local or global statistics, without revealing to anyone.

The experiments demonstrated that this additional protection can be achieved without materially affecting statistical accuracy. BFV preserves integer-valued aggregates exactly, whereas CKKS introduces negligible numerical approximation relative to DP perturbation. Thus, the observed privacy-utility trade-offs were predominantly determined by the underlying DP mechanism's query characteristics rather than by encrypted computation. Importantly, the implementation also demonstrated the compatibility of different encrypted numerical representations with the required PD operations, including the use of integer-valued perturbation for BFV statistics. The performance evaluation demonstrates that HE and DP can be integrated within a distributed healthcare analytical framework without prohibitive computational overhead for the evaluated workloads. The results support the practical feasibility of incorporating encrypted-noise processing while preserving analytical utility.

The combined HE and CDP evaluation demonstrates how complementary mechanisms can protect different stages of distributed statistical computation under the evaluated trust assumptions. However, this analytical pathway does not address collaborative predictive model training, in which information is exchanged via model updates rather than via statistical query contributions. The following chapter, therefore, examines a separate federated learning configuration and the implications of introducing patient-level DP into its training process.

8 Evaluation of Differentially Private Federated Learning

The previous evaluations addressed privacy-preserving statistical analytics, where the principal objective was to compute and release aggregate information. Collaborative predictive modelling presents a different set of requirements, as model parameters must be learnt from patient-level observations distributed across participating hospitals. This chapter, therefore , investigates whether federated learning can support useful clinical prediction under patient-level differential privacy and how the resulting performance varies across patients and participating institutions.

The experiments follow a sequence of design decisions rather than evaluating DP training as a single configuration. Initially, feature reduction is examined to determine whether a lower-dimensional clinical representation can retain sufficient predictive information. Non-private centralised and federated models are then compared to establish the performance differences associated with distributed training before introducing privacy constraints.

The subsequent experiments investigate how patient-level contributions should be defined when individuals have multiple ICU stays, and how gradient clipping and privacy budget selection affect predictive performance. Finally, the evaluation assesses whether changes in overall model utility are consistently reflected across demographic subgroups and hospital populations.

This progression separates the effects of data representation, federated optimisation, and differential privacy, while examining the wider consequences of these design choices for collaborative healthcare prediction.

8.1 Can Clinical Feature Dimensionality be Reduced Without Substantial Loss of Predictive Performance?

The initial experiments evaluated the effect of reducing the high-dimensional clinical feature space before introducing federated learning and DP. Three configurations were examined using the XGBoost evaluation procedure as in [42]: the original feature representation (`A0`), medication reduction (`A1`), and subsequent laboratory feature reduction (`A2`).

The original representation contained 5,721 predictors, including 5,222 sparse medication indicators. Grouping medication into 192 ATC3 [41] categories reduced the feature to 691 predictors. Subsequently, laboratory variables with less than 5% observation coverage were excluded, reducing the laboratory representation from 458 to 120 variables and yielding 353 predictors. The resulting `A2` representation comprised 353 predictors, including 192 ATC3 medication features, 120 laboratory variables, and 41 additional numerical clinical features. Following the removal of two constant medication indicators, the final representation contained 351 predictors and was adopted for subsequent centralised and federated learning experiments. Overall the two stages reduced the feature dimensionality by approximately 93.8%.

Table 11 presents the corresponding predictive performance on the heal-out evaluation data.

Table 11: Predictive performance across feature-reduction configurations.

Configuration	Features	ROC-AUC	PR-AUC	Accuracy	Recall	F1
A0	5721	0.8791	0.7858	0.8312	0.6303	0.6880
A1	691	0.8700	0.7737	0.8225	0.6014	0.6668
A2	353	0.8691	0.7728	0.8222	0.6007	0.6662

Medication reduction resulted in a modest decrease in overall discrimination, with ROC-AUC declining from 0.8791 to 0.8700. However, recall for prolonged ICU stays decreased from 0.6303 to 0.6014, indicating that the identification of positive cases had a greater effect than suggested by aggregate accuracy alone. In contrast, subsequent laboratory reduction produced minimal additional changes in predictive performance. ROC-AUC decreased by only 0.0009, while prolonged-stay recall remained nearly unchanged. This suggests that, for the evaluated prediction task, much of the information retained in the low-coverage laboratory variables contributed little additional predictive value beyond what was provided by the reduced representation.

These findings supported the adoption of the reduced feature space for subsequent learning experiments.

8.2 How much Predictive Performance Changes when Training is Federated?

Following feature reduction, the experiments evaluated the effect of federated training on predictive performance independently of differential privacy. A centralised logistic regression model (`F0`), trained using pooled training data, was compared with a non-private federated implementation using FedAvg across five patient-disjoint hospitals. Both configurations employed the same 351-feature representation and were evaluated on the common held-out test population.

Table 12 presents the predictive performance of the two configurations.

Table 12: Predictive performance of centralised and non-private federated learning baselines.

Metric	Centralized (F0)	Federated (F1)
ROC-AUC	0.8480	0.8414
PR-AUC	0.7413	0.7304
Accuracy	0.8101	0.8073
Precision	0.7419	0.7441
Recall	0.5628	0.5454
F1-score	0.6400	0.6294

Federated training achieved predictive performance close to the centralised reference. ROC-AUC decreased by approximately 0.0066, while accuracy declined by less than 0.3%. A larger difference was observed in prolonged-stay recall, which decreased by approximately 1.7%, indicating that the federated model identified slightly fewer positive cases.

These differences arise from the federated optimisation procedure, in which hospitals perform local model updates before parameter aggregation, rather than optimising directly over the pooled training data. Unequal hospital populations and the infinite number of communication rounds may further influence model convergence. Nevertheless, the result indicates that collaborative training across the simulated hospitals retained most of the centralised model's predictive performance without requiring consolidation of patient-level training records.

The non-private federated model, therefore, provides an essential reference for subsequent DP experiments, allowing performance changes associated with patient-level contribution constraints, gradient clipping, and noise perturbation to be explained separately from those already present in federated learning.

8.3 How can Multiple Patient-level Contributions be Represented?

The federated experiments adopted the patient as the unit of privacy, rather than an individual ICU admission. This distinction is significant because patients may contribute multiple stays to the training dataset. Consequently, the addition or removal of one patient must account for all retained observations associated with that patient.

Two patient-level strategies were considered following preliminary comparisons of single-stay, bounded multi-stay, and unrestricted representations. To inform the contribution limit, we first examined the distribution of ICU stays per patient. The analysis showed that 78.08% of patients had one stay, 14.47% had two stays, and only 7.45% had three or more stays, with a maximum of 37. Based on this distribution, Strategy A retained the earliest ICU stay per patient, whereas Strategy $B5$ incorporated up to five stays. The five-stay limit was selected as a practical trade-off between preserving longitudinal clinical information and limiting the number of observations processed per patient. This configuration retained 58,228 of the 59,522 available training stays (97.8%), while preliminary non-private comparisons indicated that predictive performance remained close to the unrestricted representation. Accordingly, $B5$ was adopted for subsequent experiments as a bounded multi-stay configuration rather than an empirically optimal contribution limit.

For patient i with k_i retained stays, the patient-level gradient and its clipped representation are defined as:

$$g_i = \frac{1}{k_i} \sum_{j=1}^{k_i} \nabla_\theta \ell(\theta; x_{ij}, y_{ij}), \qquad \bar{g}_i = g_i \min\left(1, \frac{C}{\|g_i\|_2}\right), \tag{7}$$

where $k_i = 1$ for Strategy A and $k_i \leq 5$ for Strategy $B5$. The clipping operation bounds each sampled patient's contribution independently of the number of retained stays. To isolate the influence of patient representation, both strategies were initially evaluated without gradient clipping or DP perturbation.

Table 13 presents their non-private predictive performance.

Table 13: Non-private patient-level contribution strategy comparison.

Metric	**Strategy** A	**Strategy** $B5$
ROC-AUC	0.8383	0.8388
PR-AUC	0.7193	0.7197
Accuracy	0.8046	0.8047
Recall	0.5619	0.5652
F1-score	0.6331	0.6346

Both strategies achieved comparable predictive performance, with $B5$ showing marginally higher discrimination and positive-case recall. These results indicate that retaining additional ICU stays provided a small predictive benefit under the evaluated configuration, rather than a substantial improvement. Nevertheless, $B5$ enabled the inclusion of richer patient histories while maintaining a consistent definition of patient-level contribution.

Differential privacy was subsequently incorporated at each participating hospital through patient-level Poisson sampling, gradient clipping, and Gaussian perturbation. The noise multiplier was calibrated separately for each hospital according to its sampling rate and total optimisation steps. Privacy expenditure was tracked using Rényi Differential Privacy (RDP) accounting across 30 communication rounds, with a common target $\delta = 10^{-5}$.

This configuration enabled the evaluation of patient-level privacy under different contribution strategies while accounting for the cumulative effect of repeated federated optimisation. The resulting privacy-utility trade-offs, including the separate influences of clipping and noise perturbation, are examined in the following section.

8.4 Which Factors Determine the Privacy-utility Trade-off in Federated Training?

The privacy-utility evaluation examined how patient-level gradient clipping and DP perturbation influenced predictive performance across the two contribution strategies. To distinguish these effects, clipping-only experiments were first conducted without Gaussian noise, followed by DP training under differential privacy budgets. This separation is important because performance degradation under DP-SGD may arise from both the restriction of gradient magnitudes and the stochastic perturbation required for privacy.

8.4.1 How Does Gradient Clipping Affect Performance Interdependently of DP?

Gradient clipping limits each patient's gradient magnitude to a predefined threshold C by scaling gradients whose L_2 norm exceeds this bound, while leaving smaller gradients unchanged. In DP-SGD, this operation bounds individual contributions and enables the addition of calibrated noise, although clipping alone does not provide DP.

The clipping-only experiments evaluated thresholds $C \in \{\, 0.5, 1, 2, 4\}$ against the corresponding unclipped, non-private controls. The results are presented in Table 14

The result demonstrates that gradient clipping affects the predictive performance even in the absence of DP noise. At $C = 4$, both strategies retained performance close to their unclipped controls. However, progressively smaller clipping thresholds produced greater reductions in discrimination and positive-case recall. The strongest clipping configuration $(C = 0.5)$ was particularly restrictive, reducing recall to 0.0011 for Strategy A and 0.1326 for $B5$.

These findings indicate that excessively restrictive clipping can substantially impair the identification of prolonged ICU stays, independently of the privacy budget. Among the evaluated thresholds, $C = 4$ provided the smallest utility loss and was therefore selected for the principal privacy-budget comparisons, while smaller thresholds were retained for sensitivity analysis.

Table 14: Effect of gradient clipping on non-private federated predictive performance.

Clipping	**ROC-AUC**		**Recall**	
C	A	$B5$	A	$B5$
Unclipped	0.8383	0.8388	0.5619	0.5652
4	0.8332	0.8352	0.5525	0.5461
2	0.8216	0.8263	0.5098	0.5042
1	0.8072	0.8104	0.3391	0.4090
0.5	0.7900	0.7936	0.0011	0.1326

8.4.2 How Do Privacy Budgets Affect Predictive Performance and Training Stability

With the clipping threshold fixed at $C = 4$, DP training was evaluated for $\varepsilon \in \{0.5, 1, 2, 4, 8\}$ and $\delta = 10^{-5}$. All configurations completed 30 federated training rounds with cumulative privacy accounting.

With the clipping threshold fixed at $C = 4$, the privacy–utility evaluation considered five privacy budgets, $\varepsilon \in \{0.5, 1, 2, 4, 8\}$, with $\delta = 10^{-5}$. Table 15 presents the predictive performance of both patient-level contribution strategies across the full privacy budget range. The results indicate relatively modest changes in overall discrimination as the privacy budget decreases, although reductions in recall and F1 are more pronounced at the strongest evaluated privacy setting, particularly for Strategy $B5$. These single-run results provide an overview of the privacy–utility behaviour, but do not establish the stability of individual configurations.

Table 15: Predictive performance across privacy budgets at $C = 4$ and $\delta = 10^{-5}$ (single-run results).

Strategy	ε	**ROC-AUC**	**PR-AUC**	**Recall**	**F1**
A	8	0.83338	0.71002	0.55185	0.62595
A	4	0.83323	0.71117	0.54940	0.62554
A	2	0.83237	0.70986	0.56364	0.63033
A	1	0.83296	0.70703	0.55674	0.62668
A	0.5	0.83073	0.70451	0.54139	0.61798
$B5$	8	0.83515	0.71506	0.54740	0.62548
$B5$	4	0.83564	0.71565	0.55185	0.62817
$B5$	2	0.83491	0.71580	0.54740	0.62603
$B5$	1	0.83451	0.71362	0.54539	0.62430
$B5$	0.5	0.83255	0.70894	0.52626	0.61238

To examine the variability associated with stochastic DP training, the configurations at $\varepsilon \in \{0.5, 2, 8\}$ were subsequently evaluated over three independent runs. Table 16 reports the corresponding mean and standard deviation, providing an additional assessment of the robustness of the observed privacy-utility trends.

Table 16: Predictive performance under patient-level DP at $C = 4$ (mean $\pm$ standard deviation over three runs).

Strategy	ε	**ROC-AUC**	**PR-AUC**	**Recall**	**F1**
A	8	0.83324 ± 0.00018	0.71000 ± 0.00012	0.54992 ± 0.00167	0.62518 ± 0.00082
A	2	0.83325 ± 0.00077	0.70998 ± 0.00035	0.55266 ± 0.01040	0.62608 ± 0.00373
A	0.5	0.82965 ± 0.00149	0.70474 ± 0.00162	0.55229 ± 0.01290	0.62302 ± 0.00564
$B5$	8	0.83528 ± 0.00011	0.71514 ± 0.00020	0.55066 ± 0.00346	0.62676 ± 0.00114
$B5$	2	0.83493 ± 0.00010	0.71512 ± 0.00065	0.54769 ± 0.00379	0.62610 ± 0.00249
$B5$	0.5	0.83251 ± 0.00036	0.70987 ± 0.00090	0.53167 ± 0.01076	0.61556 ± 0.00536

The replicated results show relatively stable predictive discrimination across the evaluated privacy budgets, with modest reductions under the strongest privacy setting. For strategy $B5$, mean ROC-AUC decreased from 0.8353 at $\varepsilon = 8$

to 0.8325 at $\varepsilon = 0.5$, while recall decreased from 0.5507 to 0.5317. Strategy A exhibited a comparable reduction in ROC-AUC, although its recall did not follow a monotonic trend.

$B5$ overall achieved slightly higher ROC-AUC and PR-AUC than A, but this advantage did not consistently extend to threshold-dependent measures. At $\varepsilon = 0.5$, $B5$ exhibited lower recall and F1 than A, illustrating that improved ranking performance does not necessarily imply better identification of positive cases.

Taken together, the experiments show that clipping calibration was a major determinant of predictive performance in the evaluated federated setting. At the selected clipping threshold, introducing stronger DP perturbation produced comparatively modest changes in overall discrimination, although some threshold-dependent measures were more sensitive. The replicated findings should be interpreted descriptively, as three independent runs provide only a limited assessment of training variability.

8.5 Does Patient-level DP Affect Performance Uniformly Across Patient Groups?

The subgroup evaluation examined whether patient-level DP affected predictive performance differently across racial groups and insurance groups. The analysis compared each DP configuration against its corresponding non-private patient-grouped control, thereby distinguishing pre-existing performance disparities from changes observed under privacy-constrained training. True-positive rate (TPR), representing recall for prolonged ICU stays, and false-positive rate (FPR) were used to assess differences in classification behaviour across subgroups.

For each metric m, the disparity was summarised using the maximum difference between eligible subgroups:

$$\Delta_m = \max_g(m_g) - \min_g(m_g), \tag{8}$$

where g denotes a subgroup. The primary analysis included groups with at least 100 held-out observations. Missing insurance values were excluded from the primary between-insurance disparity calculation. DP results were evaluated at $C = 4$ and $\varepsilon \in \{8, 2, 0.5\}$, with three independent executions per configuration.

Table 17 presents the observed TPR and FPR disparities for both contribution strategies. DP disparities were calculated from the mean subgroup performance across the three executions.

Table 17: Subgroup performance disparities under non-private and patient-level DP federated learning.

Attribute	**Strategy**	**Metric**	**Control**	$\varepsilon = 8$	$\varepsilon = 2$	$\varepsilon = 0.5$
Race	A	TPR	0.0901	0.1174	0.1293	0.1214
Race	A	FPR	0.0192	0.0331	0.0309	0.0289
Race	B5	TPR	0.0953	0.1021	0.1082	0.1511
Race	B5	FPR	0.0211	0.0328	0.0263	0.0309
Insurance	A	TPR	0.2014	0.1961	0.1795	0.1917
Insurance	A	FPR	0.0382	0.0417	0.0455	0.0494
Insurance	B5	TPR	0.1953	0.1973	0.1978	0.1937
Insurance	B5	FPR	0.0322	0.0358	0.0319	0.0393

8.5.1 How Does DP Affect Predictive Disparities Across Racial Groups?

The racial subgroup analysis identified recall differences that were already present in the non-private models and that generally increased under DP training. For strategy A, the TPR disparity increased from 0.0901 in the matched non-private control to 0.1214 at $\varepsilon = 05$. A larger change was observed for $B5$, where the corresponding disparity increased from 0.0953 to 0.1511.

Further examination of $B5$ showed that recall for the Asian subgroup decreased from 0.5238 in the non-private control to a mean of 0.4418 at $\varepsilon = 0.5$. This reduction contributed to the wider between-group TPR disparity under the strongest privacy setting evaluated. FPR disparities also increased relative to the controls, although their changes were smaller in absolute magnitude.

These findings indicate that relatively stable global predictive performance can coexist with more pronounced subgroup-level changes, particularly in identifying positive cases.

8.5.2 How Does DP Affect Predictive Disparities Across Insurance Groups?

In contrast, the insurance subgroup analysis revealed substantial TPR disparities already present in the non-private controls. Strategy A exhibited an initial disparity of 0.2014, compared with 0.1953 for $B5$. At $\varepsilon = 0.5$, these disparities were 0.1917 and 0.1937, respectively, indicating that the stronger privacy did not materially increase the existing differences in recall across insurance groups.

However, FPR disparities showed a different pattern. For strategy A, the FPR disparity increased from 0.0382 to 0.0494, while B5 exhibited an increase from 0.0322 to 0.0393 at $\varepsilon = 0.5$. Thus, changes in subgroup behaviour depended on the performance measure considered.

The results show that the effect of DP on subgroup performance was not consistent across racial and insurance groups. For racial groups, the differences in prolonged-stay recall generally increased under DP, with the largest increase observed for Strategy B5 at $\varepsilon = 0.5$. In contrast, insurance-related recall differences were already substantial in the non-private models and remained relatively stable after introducing DP, although differences in false-positive rates increased in some configurations. These findings highlight that preserving overall predictive performance does not necessarily preserve comparable performance across patient groups. They also demonstrate the importance of evaluating DP models against matched non-private baselines to distinguish existing disparities from changes observed under privacy-constrained training.

8.6 Do Common Privacy Targets Produce Equivalent training Conditions Across Hospitals?

The preceding evaluation examined predictive utility and subgroup disparities across the combined test population. However, the aggregate performance may conceal differences between participating hospitals. Therefore, a hospital-level analysis was conducted to examine how privacy calibration, predictive performance, and subgroup disparities varied across the five simulated hospitals.

The analysis compared matched non-private controls against DP configurations at $C = 4$, $\delta = 10^{-5}$, and $\varepsilon \in \{8, 2, 0.5\}$. Each DP configuration was evaluated over three independent executions, with performance measured separately on each hospital's held-out test observations.

8.6.1 How Does Privacy Calibration Vary between Participating Hospitals?

Although all hospitals were assigned the same target privacy guarantee, their patient populations differed substantially. Consequently, the patient-level Poisson sampling probabilities and calibrated Gaussian noise multipliers varied across hospitals.

Table 18 presents the hospital-specific training populations and privacy parameters.

Table 18: Hospital-specific patient sampling and Gaussian noise calibration.

Hospital	Patients	q_h	σ_h ($\varepsilon = 8$)	σ_h ($\varepsilon = 2$)	σ_h ($\varepsilon = 0.5$)
$\mathtt{H}_1$	8,690	0.02941	0.9154	2.1936	7.3047
$\mathtt{H}_2$	8,690	0.02941	0.9154	2.1936	7.3047
$\mathtt{H}_3$	8,690	0.02941	0.9154	2.1936	7.3047
$\mathtt{H}_4$	2,172	0.11111	1.4146	4.0881	14.1406
$\mathtt{H}_5$	15,208	0.01667	0.7943	1.7133	5.5176

$\mathtt{H}_4$, the smallest hospital, had the highest patient-sampling probability and required the largest noise multiplier to satisfy the common cumulative privacy target. Conversely, $\mathtt{H}_5$ had the largest training population, the lowest sampling probability, and the smallest calibrated noise multiplier.

These differences demonstrate that enforcing an equivalent privacy guarantee across participating hospitals does not imply identical local training conditions. The resulting variation in model performance was therefore examined using hospital-specific test populations.

8.7 Does the effect of DP on Predictive Performance Differ Between Hospitals

To assess differences in predictive utility, the performance of each DP configuration was compared against its corresponding hospital-level non-private control. Table 19 reports the mean changes at the strongest evaluated privacy setting, $\varepsilon = 0.5$

Table 19: Hospital-level predictive performance changes relative to matched non-private controls at $\varepsilon = 0.5$ and $C = 4$. Values represent DP mean minus control.

Strategy	Hospital	Δ ROC	Δ PR	Δ Recall	Δ F1	Δ FPR
A	$\mathtt{H}_1$	-0.00810	-0.01272	-0.00696	-0.00626	+0.00108
A	$\mathtt{H}_2$	-0.01045	-0.01587	-0.00683	-0.00634	+0.00111
A	$\mathtt{H}_3$	-0.00706	-0.01621	-0.01302	-0.01532	+0.00615
A	$\mathtt{H}_4$	-0.00915	-0.01049	+0.00560	-0.00374	+0.01012
A	$\mathtt{H}_5$	-0.00896	-0.01439	-0.01293	-0.01232	+0.00283
$B5$	$\mathtt{H}_1$	-0.00560	-0.00583	-0.03443	-0.01963	-0.00804
$B5$	$\mathtt{H}_2$	-0.00684	-0.01278	-0.03227	-0.01562	-0.01226
$B5$	$\mathtt{H}_3$	-0.00613	-0.01085	-0.03720	-0.02650	-0.00389
$B5$	$\mathtt{H}_4$	-0.00596	-0.00718	-0.01821	-0.00933	-0.00633
$B5$	$\mathtt{H}_5$	-0.00649	-0.01001	-0.03393	-0.01792	-0.01023

The results show that changes in predictive performance are not consistent across hospitals. Under strategy A, the largest ROC-AUC reduction occurred in $\mathtt{H}_2$ in PR-AUC and F1-score. Under $B5$, $\mathtt{H}_3$ again showed the largest F1 reduction, while $\mathtt{H}_4$ experienced the smallest F1.

Notably, $\mathtt{H}_4$ did not consistently exhibit the greatest performance degradation despite having the highest sampling probability and Gaussian noise multiplier. This indicates that hospital size and noise calibration alone are insufficient to explain differences in observed predictive utility.

A further distinction emerged in threshold-dependent performance. Under $B5$ at $\varepsilon = 0.5$, recall and FPR decreased across all five hospitals relative to their non-private controls. This indicates that the DP model identified fewer prolonged-stay cases and generated fewer false positives. The pattern was observed throughout the federation rather than being restricted to a single hospital.

8.7.1 Are Subgroup Performance changes Consistent Across Hospitals?

The hospital-level evaluation also examined whether changes in racial and insurance-related recall disparities were consistent across participating hospitals. As in the preceding subgroup evaluation, disparity was measured using the maximum difference in TPR between eligible groups.

Table 20 reports the changes in within-hospital TPR disparities at $\varepsilon = 0.5$, relative to the corresponding non-private controls. Positive values indicate an increase in disparity, whereas negative values indicate a decrease.

Table 20: Changes in within-hospital TPR disparity under DP at $\varepsilon = 0.5$ relative to matched non-private controls.

	Race		Insurance	
Hospital	A	$B5$	A	$B5$
$\mathtt{H}_1$	+0.01011	+0.02996	+0.01131	-0.00794
$\mathtt{H}_2$	+0.00944	-0.00167	-0.00803	-0.00853
$\mathtt{H}_3$	-0.00228	+0.01789	+0.01062	+0.01924
$\mathtt{H}_4$	-0.01912	+0.02118	-0.01587	+0.00529
$\mathtt{H}_5$	+0.04001	+0.04619	-0.03431	-0.03041

The racial subgroup results reveal substantial variation across hospitals. $\mathtt{H}_5$ exhibited the largest increase in TPR disparity under both configuration strategies. For $B5$, its disparity increased from 0.2058 in the non-private control to 0.2520 under DP. This occurred despite $\mathtt{H}_5$ having the largest training population and the smallest noise multiplier.

In contrast, insurance-related TPR disparities exhibited both increases and decreases. For example, under $B5$, $\mathtt{H}_5$ experienced an increase of 0.0192, whereas $\mathtt{H}_5$ showed a decrease of 0.0304. The latter hospital nevertheless retained a substantial insurance-related disparity, indicating that a reduction under DP should not necessarily be interpreted as eliminating pre-existing performance differences.

These comparisons must also account for subgroup representation. Only two racial groups in $\mathtt{H}_4$ satisfied the minimum eligibility threshold, compared with five in $\mathtt{H}_5$. Consequently, differences in the number and composition of eligible groups can affect the interpretation of maximum-minimum disparity measures.

Overall, the hospital-level evaluation demonstrates that common patient-level privacy targets do not produce uniform predictive utility for subgroup performance across participating hospitals. Although hospital-level specific sampling probabilities and noise multipliers contribute to different optimisation conditions, the observed outcomes cannot be explained by these parameters alone. The findings suggest that local data composition and classification behaviour must also be considered when assessing privacy-protecting federated learning. As the analysis involves only five simulated hospitals, these observations remain descriptive and do not establish causal relationships between hospital characteristics and performance.

8.8 Discussion and Lessons Learned: What Design Considerations Emerge from Privacy-preserving Federated Learning?

The experimental findings demonstrate that the privacy-utility trade-off in federated learning cannot be characterised solely by the selected privacy budget. Predictive performance was influenced by several aspects of the training configuration, including federated optimisation, patient-level contribution strategies, gradient clipping, and DP perturbation. Importantly, the clipping-only experiments show that restricting gradient magnitudes can substantially affect model performance even in the absence of added noise. This distinction highlights the importance of separating the effects of clipping and perturbation when evaluating the cost-utility cost of DP.

The comparison between patient-level contribution strategies further illustrates the role of representation in privacy-preserving learning. Although incorporating multiple ICU stays through strategy $B5$ provided a modest improvement in predictive discrimination, this advantage did not consistently translate into higher recall or F1-score under stronger privacy settings. Nevertheless, the strategy demonstrated how additional longitudinal observation could be incorporated through within-patient gradient aggregation while retaining the patient as the privacy unit. The results, therefore, support evaluating patient representation alongside privacy parameters rather than assuming that additional observations will necessarily improve predictive performance.

At the selected clipping threshold, overall ROC-AUC and PR-AUC remained relatively stable across several evaluated privacy budgets, with more noticeable degradation under the strongest privacy setting. However, subgroup analysis revealed that changes in global predictive performance did not fully capture differences observed in individual patient populations. Racial recall disparities increased in some configurations, particularly for $B5$ at $\varepsilon = 0.5$, whereas substantial insurance-related disparities were already present in the non-private controls. This demonstrates the need to assess both aggregate utility and subgroup-level performance when interpreting the consequences of introducing DP.

The hospital-level evaluation provided further evidence that equivalent privacy guarantees do not imply equivalent utility outcomes across participating institutions. Differences in patient populations and sampling probabilities necessitated hospital-specific noise calibration, yet hospitals that used larger noise multipliers did not consistently experience greater predictive degradation. These observations suggest that the relationship between privacy calibration and model behaviour also depends on the characteristics of local data and the resulting classifier decisions.

From a practical perspective, the findings emphasise the importance of evaluating patient-contribution definitions, clipping thresholds, privacy budgets, and predictive performance jointly when designing differentially private federated learning systems. In multi-hospital healthcare applications, this assessment should extend beyond overall accuracy and discrimination to include clinically relevant measures such as recall, false-positive rates, and subgroup differences. Privacy parameter selection should therefore be informed by the intended analytical task and its utility requirements rather than relying on a particular ε value as a universal configuration.

8.8.1 Limitations

Several limitations should be considered when interpreting these findings. First, the evaluation used five simulated hospitals constructed from a single MIMIC-IV dataset. Although the patients introduce differences in patient population and local sampling conditions, they do not capture the full institutional, demographic, and clinical heterogeneity of independently operated hospitals.

Second, the experiments focused on a fixed feature representation, a logistic regression model, and a predefined federated training configuration. Consequently, the observed privacy-utility relationships may differ for more complex models, alternative prediction tasks, or other optimisation settings. The three independent DP executions also provided a limited assessment of stochastic variability and do not establish the statistical significance of the observed differences.

Finally, the fairness analysis was restricted to racial and insurance categories, with minimum subgroup eligibility requirements. Differences in subgroup sizes and representation may influence the reported disparity measures, particularly within smaller hospitals. Furthermore, the privacy accounting addresses the patient-level DP training mechanism; extending the privacy claim to the complete analytical pipeline would require accounting for any additional data-dependent processing that is not independently privacy-preserving.

Accordingly, the results should be interpreted as evidence of the practical interactions between patient-level privacy, model utility, and subgroup performance within the evaluated federated setting, rather than as a universally optimal configuration for healthcare deployments.

8.9 Chapter Summary

This chapter evaluated the integration of patient-level differential privacy into federated learning for predicting prolonged ICU stays using MIMIC-IV data. The evaluation considered feature dimensionality reduction, centralised and federated learning baselines, patient-level contribution strategies, privacy–utility trade-offs, and subgroup performance across five simulated hospitals.

The results demonstrated that federated learning achieved predictive performance reasonably close to the centralised baseline, while patient-level DP could be incorporated without substantial degradation in overall discrimination under several evaluated configurations. However, the experiments also established that predictive utility depended on more than the selected privacy budget. In particular, aggressive gradient clipping caused considerable performance degradation even in the absence of DP noise, highlighting the importance of clipping calibration.

The comparison of patient contribution strategies showed that incorporating multiple ICU stays provided modest improvements in predictive discrimination but did not consistently improve recall or F1-score. Furthermore, subgroup evaluations revealed that relatively stable overall performance could coexist with changes in racial and insurance-related disparities. The hospital-level analysis also showed that a common privacy target required different local noise calibrations and did not produce uniform utility outcomes across participating hospitals.

Collectively, these findings demonstrate the feasibility of patient-level differentially private federated learning in the evaluated healthcare prediction setting, while highlighting the need to jointly consider patient representation, clipping parameters, privacy expenditure, and subgroup performance when assessing privacy-preserving model configurations.

9 Discussion, Limitations, and Future Directions

This chapter brings together the findings from the experimental evaluation of the proposed privacy-preserving healthcare analytics framework. It discusses the complementary roles of Differential Privacy (DP), Homomorphic Encryption (HE), Federated Learning (FL), and privacy governance across the two analytical workflows, examining their implications for privacy protection, analytical utility, and practical deployment. The chapter further considers the limitations of the implemented framework and experimental setting, before identifying directions for future research.

9.1 Synthesis of Experimental Findings

This work investigated the integration of Complementary Privacy-Enhancing Technologies (PETs) within two healthcare analytical workflows: distributed statistical analysis and federated predictive modelling. The experimental evaluation examined how differential privacy, homomorphic encryption, federated learning, and privacy governance address distinct information-exposure risks while introducing different analytical and operational requirements. Rather than treating these technologies as interchangeable privacy solutions, the framework considered their application according to the information being processed, the intended analytical output, and the associated trust boundaries.

A central observation is that privacy parameters alone are insufficient to characterise the practical utility of privacy-preserving analytics. The statistical experiments demonstrated that the same privacy budget could yield substantially different error levels depending on the statistical operation and the underlying population. Rare-subgroup estimates were particularly sensitive to perturbation, while incomplete data availability introduced additional error independently of the DP mechanism. The disclosure-risk experiments further illustrated that inference success depends on adversarial knowledge and the information available through related statistical releases. These observations highlight the importance of examining privacy guarantees alongside the characteristics of the data, analytical workload, and disclosure scenario.

The evaluation also demonstrated the complementary roles of cryptographic protection and differential privacy in distributed statistical processing. Homomorphic encryption enabled the aggregation of hospital-level statistics without exposing their plaintext values to the analytical coordinator, while incorporating encrypted DP noise prior to decryption

restricted the release service to processing perturbed aggregates within the evaluated workflow. The experiments confirmed the numerical correctness of the supported encrypted operations and quantified the additional processing overhead. Alongside these mechanisms, privacy-budget accounting, query authorisation, and audit records provided operational controls over repeated statistical releases. Together, these components demonstrated how distinct protection mechanisms can be incorporated within a coordinated analytical process, subject to defined trust and implementation assumptions.

The federated learning experiments revealed a related distinction between formal privacy protection and predictive utility. Although federated training retained performance close to the centralised reference, subsequent DP experiments showed that model behaviour depended on the patient contribution strategy, clipping threshold, and noise configuration. In particular, clipping introduced measurable performance changes independently of DP perturbation. Furthermore, relatively stable overall predictive discrimination did not necessarily correspond to stable subgroup performance. Differences in recall across racial and insurance groups, together with variations in hospital-level outcomes, demonstrated that the effects of privacy-constrained training were not uniform across the evaluated patient populations.

Taken together, the findings support the technical feasibility of integrating complementary PETs into the evaluated healthcare analytical workflows. However, the experiments also demonstrate that formal privacy guarantees must be considered alongside the analytical task, data characteristics, computational requirements, and operational trust assumptions. The framework, therefore, provides an experimental basis for examining how different privacy safeguards can be combined and evaluated within defined processing environments, rather than establishing a universally applicable privacy configuration or deployment architecture.

9.2 Complementary Privacy Protection Across Analytical Workflows

The integration of multiple PETs within the proposed framework reflects the distinct privacy requirements of statistical analytics and federated model training. Although both workflows operate on distributed healthcare data, they expose different types of information and consequently require different protection mechanisms. The experimental implementation demonstrates how these mechanisms can be incorporated at specific stages of processing, rather than assuming that a single technology provides protection throughout the analytical lifecycle.

In the statistical analytics pathway, CDP addresses disclosure through released query results but does not prevent authorised processing components from accessing exact intermediate statistics. The HE-enabled configuration introduces an additional confidentiality boundary by allowing hospital-level contributions to be encrypted before aggregation. Consequently, the analytical coordinator can perform supported operations without accessing the underlying plaintext values. Incorporating encrypted DP noise before decryption further changes the release boundary, allowing the trusted release service to process perturbed aggregates rather than exact global statistics. The resulting architecture, therefore, separates protection during computation from protection of the final statistical release, with HE and DP addressing different exposure risks.

However, this separation does not eliminate the need for trusted system components. The release services retain the private decryption capability, while the confidentiality of intermediate values depends on the appropriate key management and enforcement of the authorised processing sequence. Furthermore, the cryptographic protection of aggregates does not independently constrain information disclosed through the final result. These considerations demonstrate that combining HE and DP alters the information accessible to individual components, but the resulting protection still depends on the assumptions and boundaries of the implemented architecture.

The federated learning pathway presents a different protection requirement. Retaining patient-level records at participating hospitals reduces the need to transfer complete training datasets, but model updates exchanged during optimisation may still contain information about individual patients. Patient-level DP addresses this exposure by bounding and perturbing patient contributions before local model updates are incorporated into federated aggregation. Unlike the statistical pathway, where privacy expenditure is associated with individual query releases, the federated configuration accounts for repeated optimisation steps and communication rounds. This distinction demonstrates that applying DP within different analytical architectures requires mechanisms and accounting procedures aligned with their respective information flows.

An additional consideration is the role of privacy governance in statistical pathway, query authorisation, cumulative budget enforcement, and auditing regulation when privacy-consuming operations may proceed and maintain records of their execution. These controls complement the DP mechanism by providing operational enforcement of the configured privacy policy, rather than extending its mathematical guarantee. Similarly, privacy accounting in federated learning supports the evaluation of cumulative training expenditures but does not independently address every security threat associated with distributed computation.

The principal implication is that the effectiveness of a combined PET architecture depends on the correspondence between identified information-exposure risks, the mechanisms introduced to address them, and the trust assumptions under which they operate. The evaluated framework demonstrates this complementary approach across the two analytical workflows while retaining explicit boundaries on the protection provided. Consequently, the incorporation of multiple PETs should be assessed in terms of the specific risks mitigated and those that remain, rather than interpreted as providing comprehensive privacy protection by default.

9.3 Privacy, Utility and Operational Trade-offs

The experimental evaluation identified several trade-offs associated with incorporating PETs into healthcare analytical workflows. These trade-offs were not limited to the amount of perturbation introduced by DP, but also involved the characteristics of the analytical workload, the construction of privacy-preserving mechanisms, and the additional computational requirements of cryptographic processing. Consequently, their practical significance depends on the type of analysis and the protection requirements of the corresponding workflow.

In statistical analytics, the relationship between privacy expenditure and utility was strongly influenced by population sizes and query sensitivity. Although common-population counts retained high accuracy under the evaluated privacy budgets, rare-subgroup statistics exhibited substantially greater relative error. Bounded-mean queries were particularly sensitive when computed over small populations, as perturbations to both the sum and the count introduced variability into the resulting estimate. The comparison of alternative mean-release mechanisms further demonstrated that the construction of a DP query affects its utility under a fixed definition of adjacency. These results indicate that increasing the privacy budget is not the only consideration when seeking accurate statistical releases; query formulation, contribution bounds, and the available population also influence the resulting estimates.

The disclosure-risk experiments introduced an additional dimension to this trade-off. Stronger perturbation reduced the observed success of the evaluated attribute-inference and differencing attacks, but also affected the accuracy of released statistics. Conversely, weaker perturbations generally improved statistical utility while providing less resistance to the specific inference attacks considered. The experiments also demonstrated that the relationship between statistical accuracy and disclosure risk depends on the adversary's auxiliary knowledge. Therefore, the suitability of a privacy configuration cannot be assessed solely by estimation error; it must also consider the disclosure scenario and the information potentially available to an adversary.

The HE-enabled evaluation revealed a different trade-off between intermediate-data confidentiality and computational performance. Encrypted aggregation preserved the numerical fidelity of the supported statistical operations, with negligible CKKS approximation error relative to DP perturbation. However, incorporating encrypted noise before decryption increased mean end-to-end query latency by approximately 24.9% compared with the earlier HE-enabled configuration in which noise was applied after decryption. Hybrid queries that require both BFV and CKKS also involve larger ciphertext representations and additional processing. These findings demonstrate that reducing plaintext exposure within the analytical workflow introduces measurable computational costs, even when the accuracy of the underlying statistical operation is largely preserved.

In federated learning, the privacy–utility relationship was influenced by both DP perturbation and the constraints introduced during optimisation. The clipping-only experiments demonstrated that restrictive gradient bounds could substantially reduce predictive recall independently of Gaussian noise. At the selected clipping threshold, overall discrimination remained comparatively stable across several privacy budgets, although stronger privacy settings affected some threshold-dependent metrics and subgroup outcomes. The comparison of patient contribution strategies further showed that retaining additional longitudinal observations did not consistently improve all predictive measures. Thus, the practical utility of patient-level DP depended on the interaction between privacy calibration, contribution representation, and the intended prediction task.

Operational governance introduced further constraints on how privacy-consuming analyses could be performed. Enforcing cumulative expenditure limits and requiring authorisation for high-cost queries restricted the execution of requests that exceeded the configured policy, while preserving the integrity of the privacy budget. These controls support accountable operation but also require decisions concerning budget allocation and the availability of future analytical requests. Unlike statistical distortion or cryptographic latency, this represents an operational trade-off between permitted analytical activity and the management of cumulative privacy expenditure.

Privacy Budget Implications for Statistical Analytics. The experimental findings provides a basis for identifying candidate privacy configurations according to the analytical workload. For large population counts and bounded means, ε values between 0.1 and 0.5 represent a reasonable initial setting for further evaluation, as these configurations retained high statistical accuracy in the examined queries. However, this recommendation does not extend directly to

small subgroups, where substantial estimation errors persisted even at $\varepsilon = 1$. Furthermore, the controlled inference experiments demonstrated increased disclosure risks at higher privacy budgets. These findings suggest that increasing ε to improve small-subgroup utility should not be the default response. Instead, query reformulation, population aggregation, and explicit utility requirements should inform the selection of an acceptable privacy configuration.

Privacy Budget Implications for Federated Learning. For the evaluated FL configuration $\varepsilon = 2$, with $\delta = 10^{-}5$ and clipping threshold $C = 4$, provides a reasonable reference setting for balancing privacy protection and predictive utility. At this configuration, predictive discrimination remained close to the results obtained at $\varepsilon = 8$, suggesting limited additional benefit from the larger privacy budget. Stronger protection at $\varepsilon = 0.5$ also preserved much of the overall predictive discrimination, although greater changes were observed in recall and subgroup disparities, particularly under the B5 contribution strategy. Therefore, $\varepsilon = 2$ may be considered a starting point for subsequent evaluation, while $\varepsilon = 0.5$ remains an alternative where stronger privacy is required and clinically relevant performance measures remain acceptable. These values represent findings from the evaluated experimental configuration rather than generally applicable privacy thresholds for healthcare deployment.

Across the evaluated workflows, the findings demonstrate that privacy protection, analytical utility, and operational feasibility cannot be optimised independently. The relevant trade-offs differ between statistical releases and predictive modelling, and even within the same workflow, they depend on the data, mechanism configuration, and required analytical outcomes. Importantly, the privacy parameters used for the central statistical queries and federated model training represent different privacy mechanisms and accounting settings; their numerical values should therefore not be treated as directly interchangeable measures of protection. The results support a context-dependent evaluation of PET configurations in which privacy guarantees are considered alongside measurable utility requirements, computational costs, and operational constraints.

9.4 Implications for Healthcare Data Analytics

The practical relevance of the proposed framework lies in translating privacy-preserving mechanisms into defined analytical procedures that healthcare organisations can assess and govern. Rather than introducing PETs with independent technical safeguards, the framework illustrates how privacy requirements can inform the design of statistical services, the allocation of computational responsibilities, and the conditions under which analytical results are made available.

For healthcare statistical services, the implementation suggests the value of establishing a controlled query environment in which permissible operations, contribution limits, and privacy expenditures are defined before execution. Such an arrangement allows organisations to associate analytical requests with explicit privacy policies and maintain records of authorised releases. In practice, this would require decisions about which analyses are permitted, how privacy budgets are allocated across research or reporting activities, and who is authorised to approve expenditures. These decisions are organisational responsibilities that cannot be resolved through noise calibration alone.

In multi-institutional analytics, participating organisations would also need consistent definitions of the populations, clinical variables, and statistical measures being analysed. Differences in coding practices, eligibility criteria, or data availability could affect the interpretation of combined results even when privacy mechanisms operate correctly. Establishing common data specifications, verifying participating data sources, and documenting the coverage of each analysis would therefore be important prerequisites for meaningful cross-institutional reporting.

The framework also highlights the need to assign responsibility for sensitive operations across the analytical infrastructure. In an HE-enabled service, cryptographic key custody, decryption authorisation, and release approval would require explicit operational controls. Similarly, collaborative model training would require participating hospitals to agree on definitions of patient contributions, training procedures, and validation requirements for shared models. These arrangements would need to be supported by institutional policies, access controls, and appropriate oversight rather than relying exclusively on the properties of the underlying technologies.

Finally, the framework provides a basis for incorporating technical privacy claims into a broader risk assessment process. For a healthcare organisation undertaking a Data Protection Impact Assessment (DPIA), the relevant considerations are whether a proposed safeguard identifies and addresses disclosure risk, under which assumptions it operates, and what responsibilities or residual risk remain. The experimental workflows illustrate how such questions can be made explicit in system design, providing a starting point for evaluating privacy-preserving analytics before considering their adoption in operational healthcare environments.

9.5 Limitations and Future Work

Several limitations define the scope within which the proposed framework and its experimental findings should be interpreted. Although the implementation demonstrates the integration of complementary PETs into the healthcare analytical workflow, the evaluation was conducted under controlled conditions that do not fully reflect operational, multi-institutional environments.

First, both analytical pathways were evaluated using MIMIC-IV data distributed across five simulated hospital nodes. While this configuration enabled controlled experimentation with distributed processing and unequal patient populations, it does not capture the organisational, infrastructural, and clinical differences that may arise between independently operated healthcare institutions. Furthermore, the containerised experimental environment does not reproduce the network conditions, system availability requirements, or administrative constraints of a geographically distributed deployment. Consequently, the findings cannot establish the framework's performance or operational reliability in a production healthcare infrastructure.

Second, the security assessment was conducted within a defined honest-but-curious threat model. The framework assumes that participating components adhere to the prescribed protocols and that trusted services fulfil their designated responsibilities. Attacks involving compromised hospital infrastructure, collusion between trusted components, or unauthorised access to cryptographic keys were outside the scope of validation. In particular, the HE-enabled architecture retains a trusted release service with decryption capability, making its protection dependent on the integrity of the corresponding trust boundary. Therefore, the experimental validation of encrypted computation and protected release does not constitute a comprehensive security assessment against malicious adversaries.

Third, the privacy guarantees are specific to the mechanisms, contribution definitions, and accounting procedures implemented in each analytical pathway. The statistical component applies DP to authorised query releases, whereas the federated component accounts for patient-level contributions during iterative training. These guarantees should not automatically be extended to every preceding data-processing operation or to information outside the defined protection boundaries. Similarly, privacy-budget enforcement and auditing provide operational controls but do not independently establish the correctness of all privacy claims or compliance with data-protection requirements. A deployment-specific assessment would therefore need to examine the complete processing lifecycle and the assumptions supporting each safeguard.

The experimental scope also limits the generalisability of the findings. Statistical analytics was evaluated using a predefined set of aggregate queries, while disclosure-risk experiments employed controlled adversarial scenarios and synthetic cohorts designed to isolate specific inference behaviours. These experiments provide evidence about the evaluated mechanisms but do not characterise the full range of possible analytical workloads or adversarial strategies. Likewise, the computational benchmarks reflect the selected implementation and hardware configuration rather than performance under large-scale concurrent operation. The model-specific and subgroup-related limitations of the federated evaluation have been considered separately in Chapter 8.

Overall, these limitations position the framework as an experimentally validated implementation of selected privacy-preserving analytical workflows under explicit assumptions, rather than a fully assessed operational healthcare platform. Establishing its suitability for wider institutional adoption would require evaluation in independently managed data environments, across a broader range of workloads, and through more extensive security and operational testing.

9.5.1 Future Research Directions

The framework provides a basis for further investigation into the integration and operational use of complementary PETs in healthcare analytics. Several directions emerge from the implementation and evaluation, particularly concerning multi-institutional deployment, analytical flexibility, security assurance, and the robustness of privacy-preserving modelling.

An important direction for future work is to evaluate the proposed framework using real-world healthcare data from NHS hospitals. While the current evaluation uses MIMIC-IV data partitioned across five simulated hospital nodes, further validation using datasets from participating NHS trusts would provide a more realistic assessment of the framework's applicability to healthcare environments. Such an evaluation would examine the effectiveness of the proposed privacy-preserving mechanisms across hospitals with different patient populations, clinical data characteristics, and analytical requirements. It would also allow the assessment of privacy guarantees, analytical utility, and computational performance under real-world institutional and operational conditions, providing evidence to inform the potential adoption of the framework within NHS healthcare analytics.

Further research could extend the framework to support a broader range of healthcare analytical queries and more complex clinical use cases beyond the predefined statistical operations evaluated in this work. For example, an oncology

application could involve analysing cancer diagnoses, disease-stage distributions, treatment patterns, and patient outcomes across participating hospitals. Other potential applications include examining hospital readmission trends, treatment effectiveness, and clinical outcomes across different patient populations. Such analyses may require more complex operations, including multi-attribute queries, longitudinal analysis, and comparisons across overlapping patient cohorts. These extensions introduce additional challenges for differential privacy, particularly in determining query sensitivity, bounding individual contributions, and managing cumulative privacy expenditure across related analyses. They would also require assessing whether the existing homomorphic encryption mechanisms can support the necessary computations or whether additional cryptographic operations are required. Beyond statistical analysis, the framework could be extended to more complex predictive applications, such as cancer risk prediction, using privacy-preserving federated learning. Evaluating these use cases would help establish the proposed framework's applicability to a wider range of healthcare tasks and examine the associated trade-offs among privacy protection, analytical utility, and computational requirements.

The HE-enabled pathway presents opportunities to strengthen the security assumptions underlying encrypted computation and protected release. Future investigations could consider distributed key management, threshold decryption, and more extensive adversarial testing to examine reliance on individual trusted services. Formal analysis of the interaction between encrypted numerical representations and DP perturbation would also strengthen assurance that the implemented release mechanism satisfies its intended privacy definition. In addition, larger workloads and concurrent analytical requests could be evaluated to characterise the computational and communication requirements of encrypted processing under different deployment conditions.

For federated learning, future research could extend the evaluation to alternative predictive models, clinical tasks, and patient contribution strategies. Particular attention should be given to the interaction between privacy calibration and model performance across heterogeneous institutions and under-represented patient populations. More extensive independent training repetitions and subgroup-level evaluations would help assess the stability of observed performance differences. Further investigation could also consider protecting additional data-dependent stages of model development and examining the confidentiality of information exchanged during federated optimisation, thereby extending the assessment beyond the current training-level DP mechanism.

Finally, future development could explore how the framework's technical outputs can be incorporated into systematic organisational privacy assessments. This could include linking documented privacy guarantees, trust assumptions, analytical utility requirements, and operational controls to the risks identified within a DPIA. Such an approach would support the evaluation of PET configurations in relation to specific healthcare processing requirements, while distinguishing experimentally demonstrated protections from assumptions and residual risks requiring additional assessment.

10 Conclusioin

Healthcare organisations increasingly rely on sensitive patient data to support clinical research, population health analysis, service planning, and predictive modelling. However, enabling these activities across healthcare institutions requires appropriate safeguards against information disclosure throughout data processing and analysis. This work addressed this challenge by designing and implementing a privacy-preserving healthcare analytics framework that integrated complementary Privacy-Enhancing Technologies (PETs) across trusts.

The framework combines Differential Privacy (DP), Homomorphic Encryption (HE), Federated Learning (FL), and privacy governance to address distinct privacy requirements across these analytical workflows. Through experimental evaluation using MIMIC-IV data distributed across five simulated hospitals, the work demonstrated the feasibility of integrating these mechanisms and examined their implications for analytical utility, computational performance, and privacy protection. The finding also highlighted the importance of aligning privacy configuration with the intended analytical task, the characteristics of healthcare data, and the trust assumptions of the processing environment.

The principal contribution of this work is an implemented framework that illustrates how healthcare organisations could incorporate complementary privacy safeguards into analytical processes without requiring the centralisation of patient-level records. Rather than proposing a universal privacy configuration, it provides a practical foundation for evaluating privacy protection alongside analytical requirements and operational controls. Ultimately, the work contributes towards the development of healthcare data analytics systems in which sensitive information can be used to generate valuable insights while maintaining clearly defined privacy protections and accountability.

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